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**FSAE Vehicle Model Lap Time Simulation Development and
Optimization**

FINAL REPORT

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Declaration

I hereby declare that this dissertation represents my own work and that it has been written by me in its entirety. I confirm that all sources of information have been acknowledged by appropriate references, and that this work has not been submitted, either in whole or in part, for any other degree or qualification at any institution.

I acknowledge the use of artificial intelligence (AI) tools in the preparation of this dissertation. AI tools, Claude from Anthropic and Gemini from Google, were used solely for language refinement and limited coding support in the development of the Final Year Project website. All substantive ideas, analysis, and conclusions presented in this work are my own, and I take full responsibility for its accuracy and integrity.

Name & Signature:

DAI BAIZHOU

A handwritten signature in black ink, appearing to read 'DAI BAIZHOU', written in a cursive style.

Date: 6th April, 2026

Acknowledgement

I would like to express my sincere gratitude to my supervisor, Mr. Lim Wong Wee and Mr. Kenneth Neo, for their invaluable guidance, patience, and continuous support throughout the course of this project. Their insights and encouragement have been instrumental in shaping both the direction and quality of this work.

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I would also like to thank my teammates in NUS Formula SAE team for their support and for providing a stimulating environment for discussion and learning. I am grateful their collaboration and shared passion for engineering, which have significantly enriched my learning experience and inspired many aspects of this work. Beyond this project, they have also been my best friends throughout my journey in NUS.

Finally, I would like to thank my family for their unwavering belief in me throughout this journey. This would not be possible without their unconditional love. I am truly grateful for everything they have done for me, and this work stands as a reflection of their support as much as my own efforts.

Abstract

This report presents the development of LTS26, the second generation of an in-house Lap Time Simulation (LTS) tool for the NUS Formula SAE Electric vehicle. Built upon a four-wheel vehicle model with three degrees of freedom, LTS26 represents a substantial advancement over its predecessor LTS25, which employed a simpler two-wheel bicycle model. The key architectural innovation of LTS26 is the decoupling of vehicle performance evaluation from track representation through the construction of a Performance Envelope, a pre-computed characterisation of the vehicle's dynamic limits across its full operating range. This approach reduces computation time by more than half relative to LTS25, while simultaneously supporting a more complex vehicle model.

The simulation framework differentiates between event types: a transient Simulink model handles the Acceleration and Braking events, while a quasi-steady-state MATLAB model addresses the Autocross and Endurance events through construction of a G-G-V diagram. Numerical optimisation using IPOPT, interfaced through CasADi for symbolic automatic differentiation, is employed to solve the nonlinear vehicle dynamics equations at each operating point. Subsystem models for aerodynamics, Ackermann steering geometry, and brake bias are integrated and evaluated for their influence on simulation fidelity and numerical stability.

Validation against telemetry data from the 2025 Formula SAE Electric Competition demonstrates close agreement in speed profiles for both straight-line and cornering events. Remaining discrepancies are attributed to powertrain thermal degradation and the inherent limitations of the quasi-steady-state assumption under transient driving conditions. Future development directions include improving optimisation convergence reliability and progressive expansion of the vehicle model towards higher degrees of freedom.

1 Introduction

1.1. NUS Formula SAE

Formula SAE (FSAE) is an international intervarsity competition organized by the Society of Automotive Engineers (SAE) International, where universities from around the world design, manufacture, and race formula-style cars. Each vehicle is evaluated through a series of dynamic and static events that assess its design, cost, and performance. All cars must strictly comply with the safety standards set by the judges, which are thoroughly inspected during the technical scrutineering process.

The National University of Singapore (NUS) FSAE Team entered the Electric Vehicle (EV) category in 2022 and, by 2025, achieved an impressive 7th-place finish among 86 teams, marking the best result in the team's history.



Figure 1 NUS FSAE R25 Team

1.2. Lap Time Simulation

Lap Time Simulation (LTS) is a computational modelling tool used to predict a vehicle's maximum on-track performance based on its design parameters, enabling teams to make data-driven design decisions. Previous members of the NUS FSAE Team had attempted to develop similar tools in past years, though none were built in a systematic or scalable manner. The most recent version of the self-developed LTS tool was authored by me from May 2024, started with a point-mass vehicle model and later evolved to a two-wheel (bicycle) vehicle model, now runs on a four-wheel model with 3 degrees of freedom.

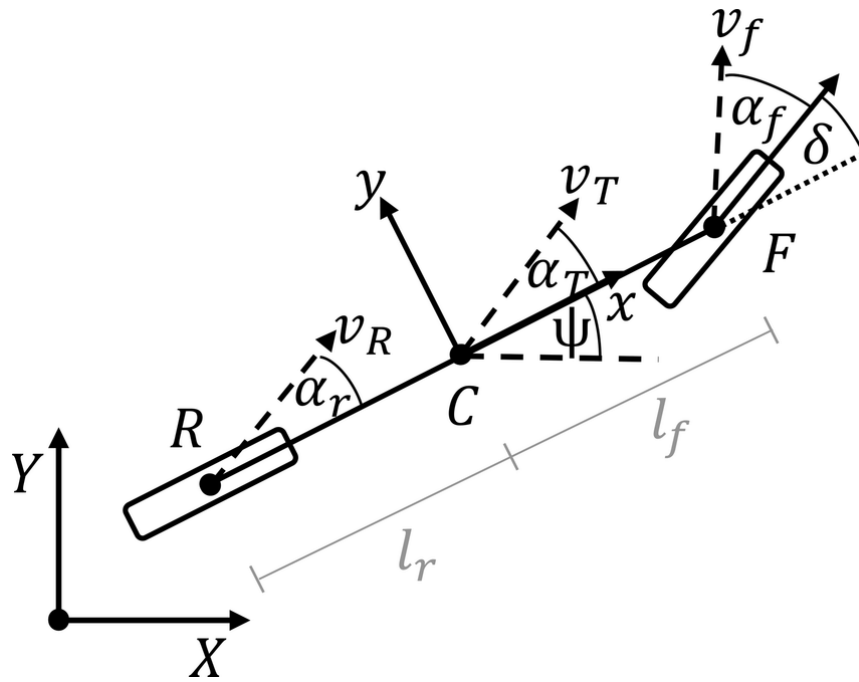


Figure 2 Vehicle Dynamics Bicycle Vehicle Model (Christoph Allig, Gerd Wanielik, 2019)

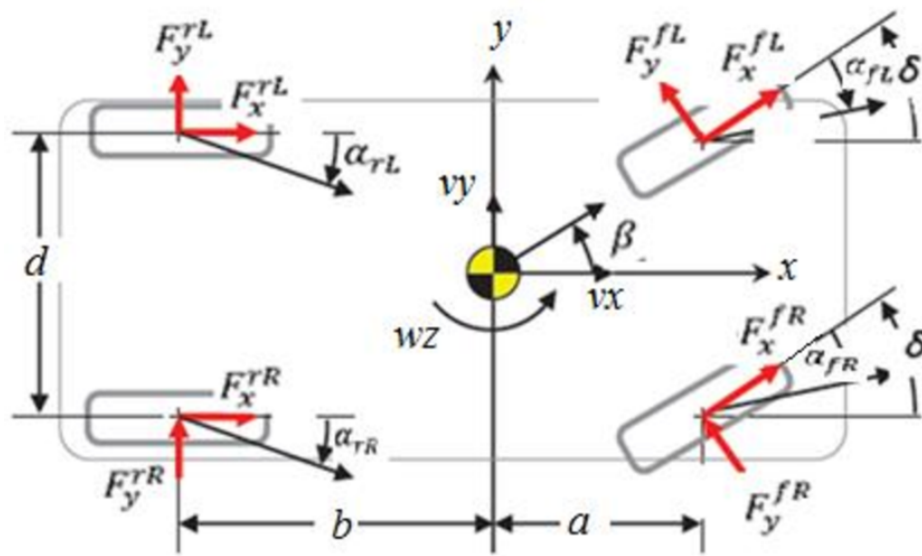


Figure 3 Vehicle Dynamics 4-Wheel Model (Ahmed, 2022)

2 Problem Definition

The development of a Lap Time Simulation tool is of considerable importance to a motorsport engineering team, as it enables detailed evaluation of vehicle performance without reliance on physical testing, some of which may be prohibitively difficult or costly to conduct. Such a tool facilitates the early identification of suboptimal design decisions, thereby reducing design iteration cycles and associated development costs. It is therefore essential for the NUS FSAE Team to maintain and continuously improve an in-house LTS tool to support race car design and setup decisions throughout the development season.

The previous iteration of the simulation tool, designated LTS25 and developed during the preceding season, was based on a two-wheel bicycle model with three degrees of freedom. Its underlying algorithm relied extensively on iterative computation, resulting in significant processing times per simulation run, with computational load scaling directly with the number of discretisation points in the track model. Furthermore, due to its inherent inability to handle multivariable problems directly, the tool required numerous simplifying assumptions, rendering it difficult to extend to more complex vehicle representations.

In addition, a single unified simulation configuration was applied across all competition events, including the Acceleration and Skidpad events, in which the vehicle operates primarily under steady-state conditions, as well as the Autocross and Endurance events, which involve considerably more complex vehicle dynamics. This approach forced the simulation to compromise accuracy in order to satisfy all use cases simultaneously, and was found to be computationally inefficient in practice.

In view of these limitations, a complete overhaul of the core simulation architecture was undertaken to improve accuracy, computational efficiency, and modularity. The resulting tool, developed under this initiative, is designated LTS26.

3 Design Statement

In the preceding season, the team directed its efforts towards improving the dynamic performance of the vehicle, ultimately achieving top-ten finishes across all dynamic competition events. While this represents a significant milestone in the team's development trajectory, considerable scope for further improvement was identified. Accordingly, the team's overarching goal for the current season was established as:

“Extract maximum performance through improved reliability”

In support of this objective, the specific goal for the Lap Time Simulation department was defined, forming the basis of the following design statement:

“Develop Lap Time Simulation software with correctness, accuracy and computational efficiency.”

4 Methodology

4.1 G-G-V Diagram

The tyre is the sole interface between the vehicle and the road surface, and consequently determines the magnitude of force that can be transmitted to generate acceleration or deceleration. The tyre's finite grip capacity may be utilised in the longitudinal direction, the lateral direction, or as a combination of both simultaneously. When the available friction is distributed between these two directions, the resultant force capability traces a circular boundary, a relationship commonly referred to as the Friction Circle.

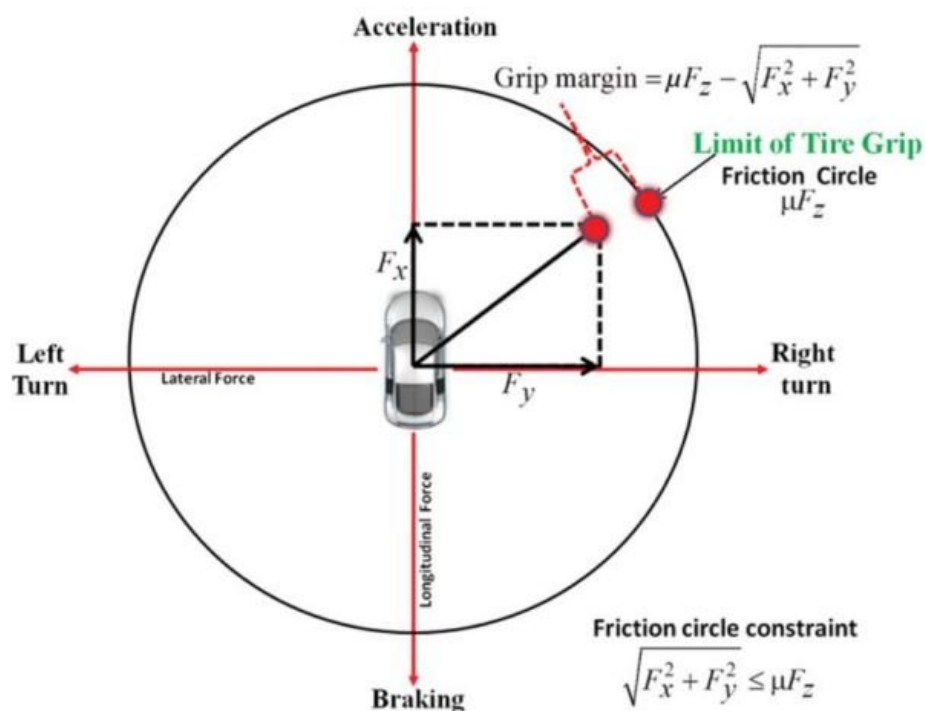


Figure 4 Racing Tires Friction Circle (Marashian, 2018)

While the Friction Circle defines the grip limits of the tyre in isolation, a race car's overall dynamic potential is further constrained by subsystems beyond the tyres alone. For an FSAE vehicle, the achievable acceleration is in certain conditions limited below the theoretical tyre grip ceiling. To translate tyre-level performance limits into actual vehicle behaviour, it is necessary to construct a G-G diagram for the vehicle as a whole. This

diagram captures the combined influence of all major vehicle subsystems to represent the car's true performance envelope under a given operating condition.

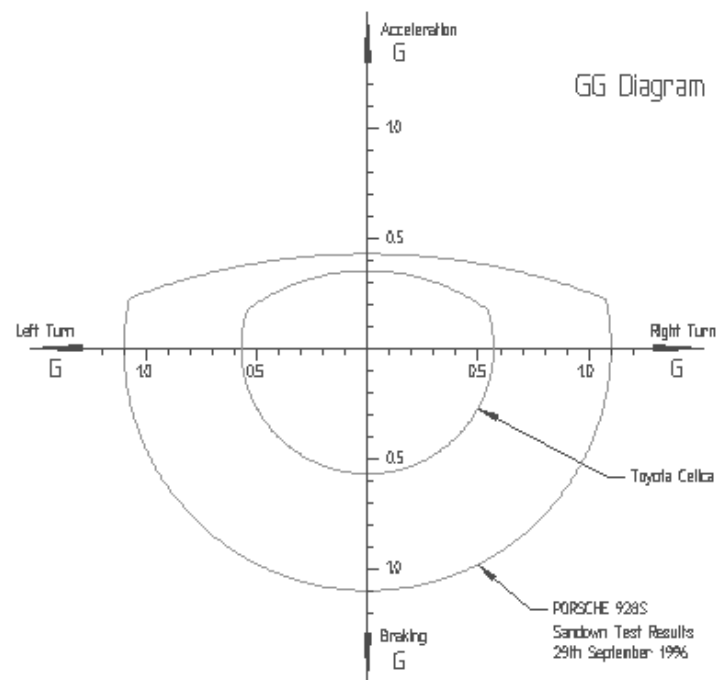


Figure 5 G-G Diagram (McIver, 1996)

Many of a vehicle's performance characteristics vary as a function of speed. Aerodynamic downforce, for instance, increases with the square of vehicle velocity, which in turn modifies the normal load acting on each tyre and consequently alters the shape of the G-G diagram. It is therefore necessary to account for the G-G diagram across the full speed range of the vehicle to obtain a complete and accurate representation of its dynamic capabilities.

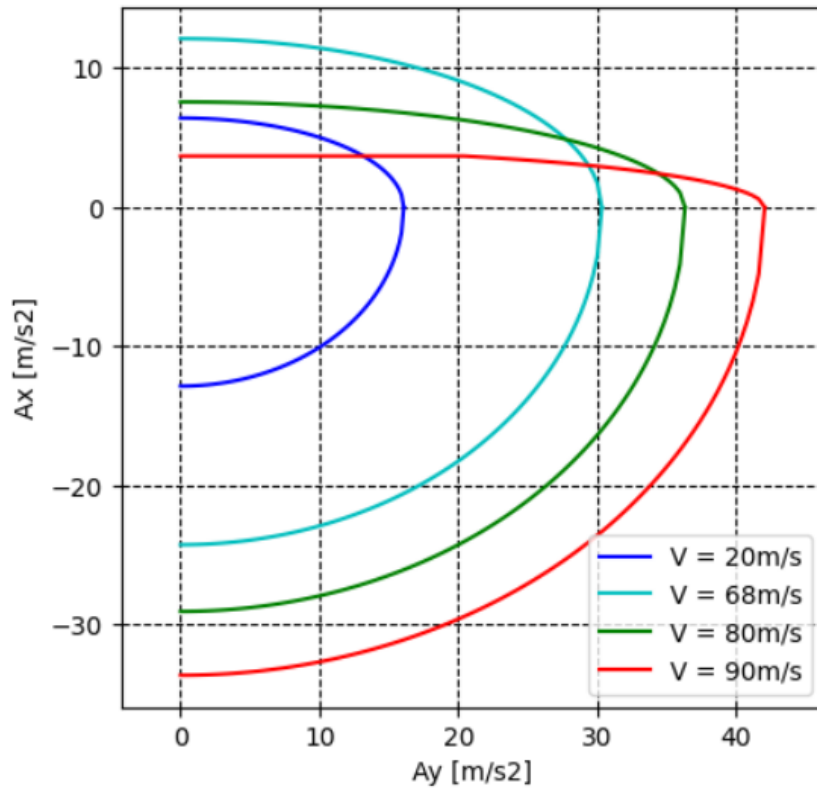


Figure 6 G-G Diagram at different speeds (Patrucco, 2025)

By constructing and combining the G-G diagrams evaluated across all speeds at which the vehicle may operate, a three-dimensional performance surface is obtained. This surface, known as the G-G-V diagram, describes the vehicle's dynamic limits comprehensively across all operating conditions. Ideally, the vehicle should operate along the boundary of this surface at all times. Any operating point that falls within the interior of the diagram indicates that the vehicle's full performance potential is not being utilised.

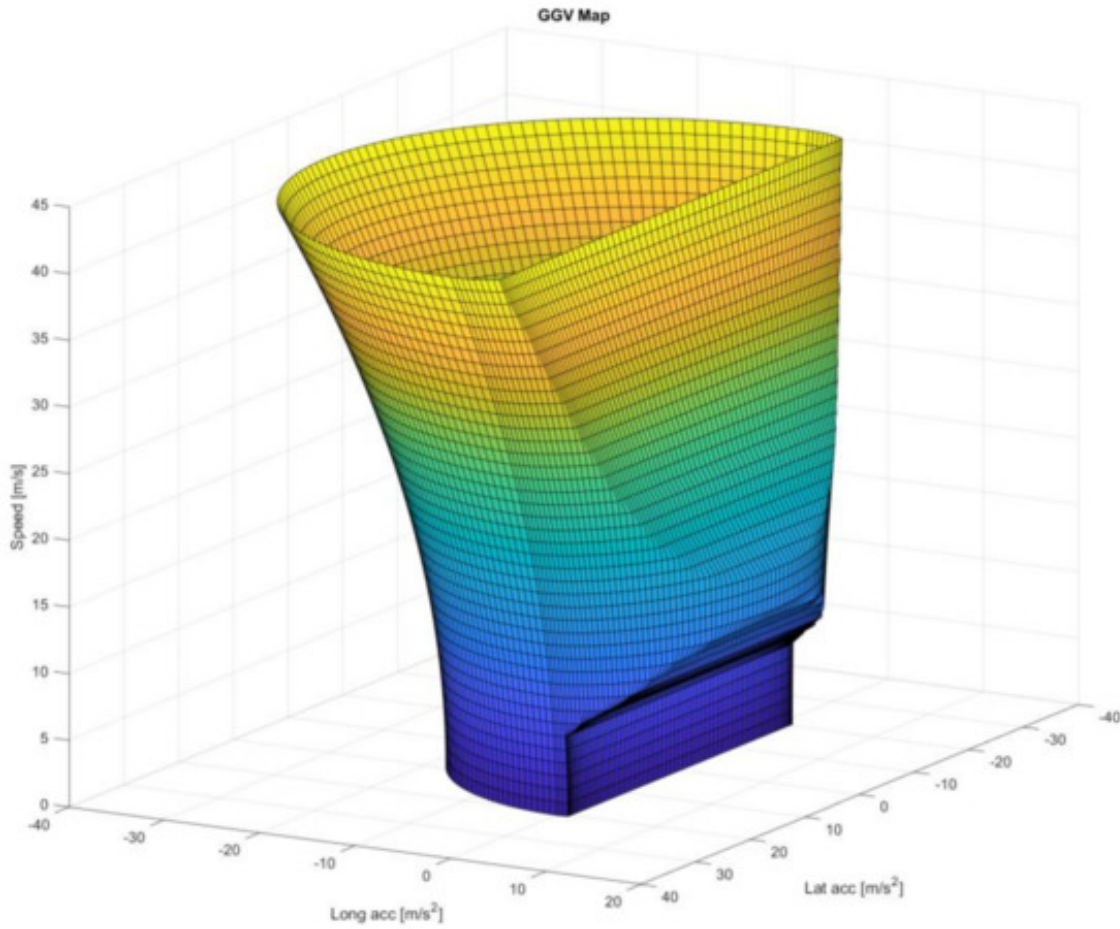


Figure 7 G-G-V Diagram (Andrei-Cristian Pridie, Csaba Antonya, 2021)

4.2 Numerical Optimization

The vehicle system is governed by a set of highly interdependent equations involving numerous unknown variables, for which closed-form analytical solutions generally do not exist. Evaluating each equation independently would be both mathematically complex and computationally inefficient. In this work, the problem is instead formulated as a numerical optimisation problem, enabling simultaneous evaluation of all decision variables within a unified framework.

The class of problems addressed here falls under nonlinear programming, which concerns the solution of optimisation problems in which either the objective function or one or more of the constraints are nonlinear in nature. In a two-dimensional context, the objective is to identify the global minimum of the cost function over a defined feasible domain. The complexity of this problem grows substantially as the number of decision variables and constraints increases.

The numerical optimisation solver selected for this work is the Interior Point OPTimiser (IPOPT), an open-source software package designed for large-scale continuous

nonlinear optimisation. IPOPT is formulated to identify local solutions to mathematical programmes with nonlinear objective functions and constraints. Relative to other non-commercial optimisation solvers, IPOPT demonstrates superior performance on nonlinear problems involving large numbers of variables, making it particularly well-suited to the vehicle dynamics application considered here.

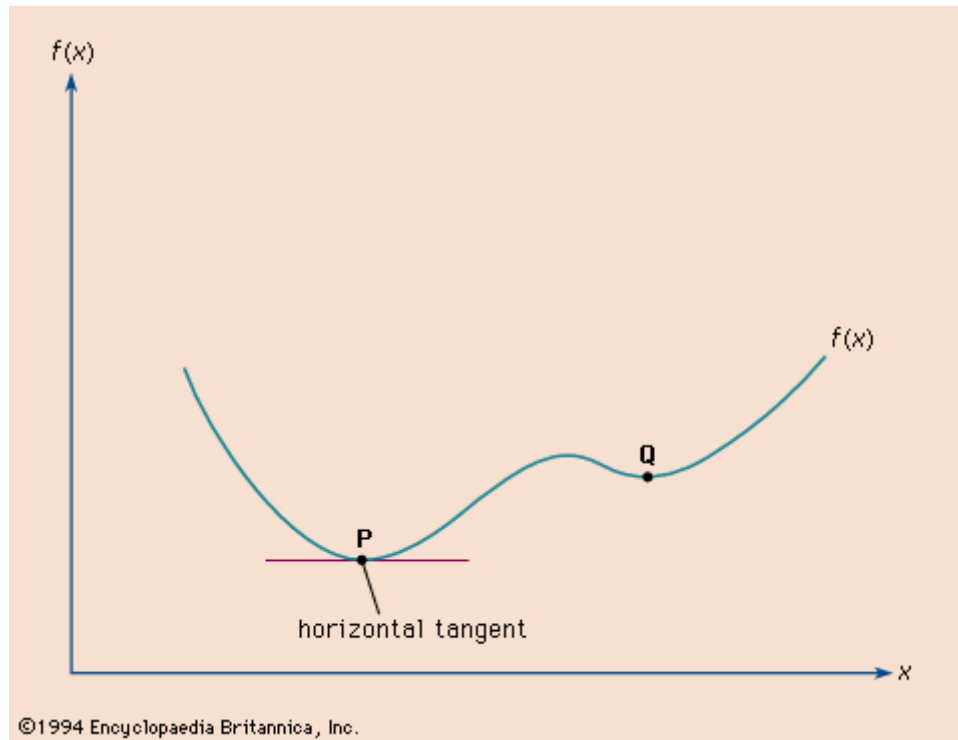


Figure 8 Visualization of non-linear optimization (Britannica, 2025)

To perform nonlinear optimisation using IPOPT, the problem must first be formally defined by specifying the decision variables, objective function, equality and inequality constraints, initial guesses, and numerical bounds. This formulation is constructed symbolically using CasADi, an open-source tool for algorithmic differentiation and numerical optimisation, which interfaces directly with IPOPT to compute the optimised decision variables and the corresponding value of the objective function.

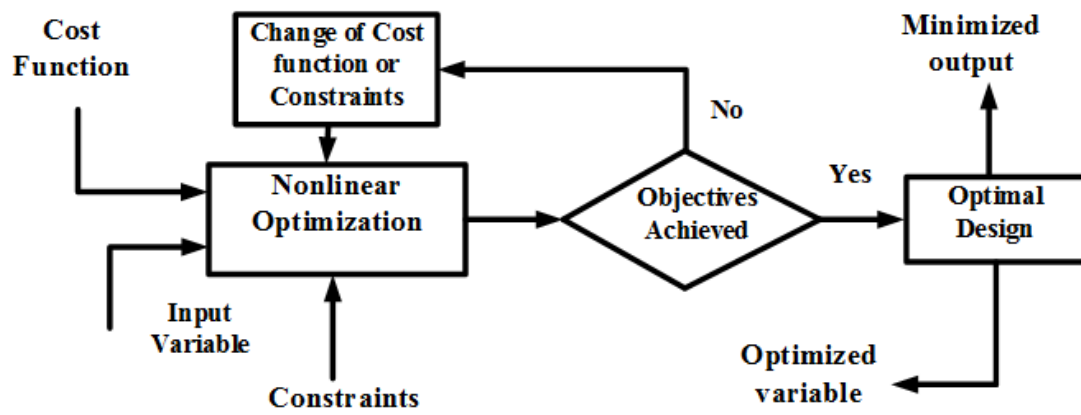


Figure 9 Functional block diagram of the nonlinear optimization methodology (Fouzia Ferdous, ABM Harun-ur Rashid, 2024)

5 Design and Prototyping

5.1 Design Overview

5.1.1 Simulation Structure Overview

LTS26 comprises over 3,000 lines of code implemented across MATLAB and Simulink, incorporating both a transient model and a quasi-steady-state model to address the distinct requirements of different competition event types. The tool produces comprehensive performance outputs for all dynamic competition events, while achieving a reduction in computation time of more than half relative to the preceding LTS25 tool.

The simulation programme is structured into two primary components: the Vehicle Model and the Dynamic Model. The Vehicle Model defines the mathematical representation of the race car's dynamics, consisting of a set of equations that describe the physical behaviour of the system. The Dynamic Model, in contrast, contains the algorithms that govern the execution of the simulation. During operation, the Dynamic Model repeatedly invokes the Vehicle Model to evaluate vehicle performance under varying conditions. In this architecture, the Vehicle Model serves as the computational engine, while the Dynamic Model governs the conditions under which this engine is exercised.

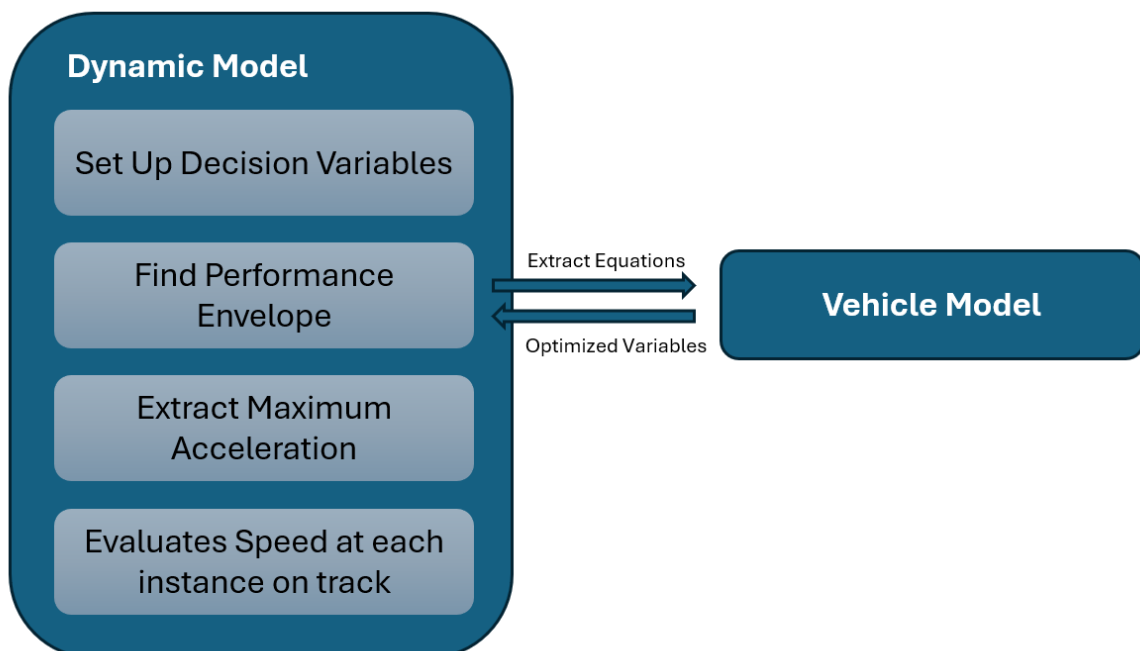


Figure 10 Dynamic Model versus Vehicle Model

5.1.2 Revamped Structure

A primary contributor to the computational inefficiency of LTS25 was that its processing time scaled directly with the number of discretisation points in the track model. At each mesh point, the simulation independently evaluated the vehicle's cornering, acceleration, and braking capabilities, with the output updated point by point as the computation progressed.

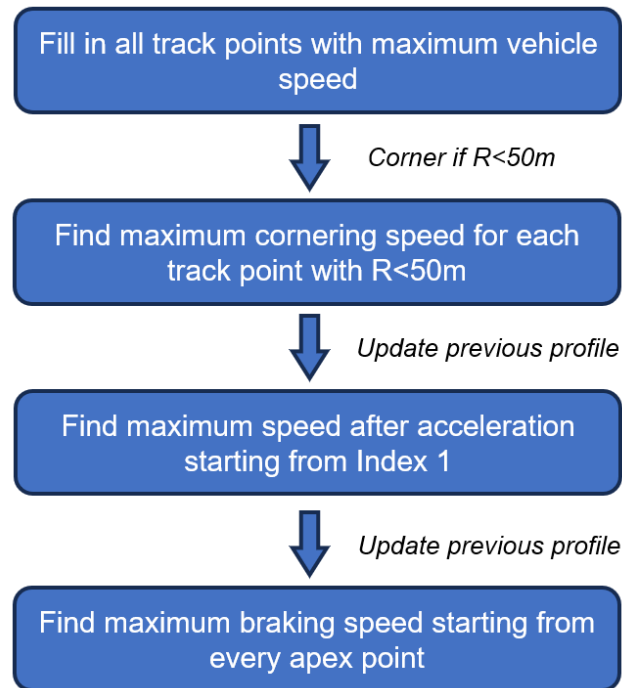


Figure 11 LTS25 Simulation Structure

To address this fundamental inefficiency, the simulation framework in LTS26 decouples vehicle performance evaluation from the track representation entirely. Rather than performing point-by-point calculations along the track, the simulation first constructs a comprehensive representation of the vehicle's performance, the Performance Envelope. This envelope characterises the vehicle's maximum capabilities across the full range of operating conditions.

Since vehicle performance varies with speed and other operating parameters, the envelope is constructed by evaluating the vehicle model at discrete operating points, followed by interpolation to generalise the results across the continuous operating domain. Once established, this Performance Envelope can be applied to any track geometry without repeating the underlying vehicle model computations, substantially

reducing the total computational cost and significantly improving the flexibility and reusability of the simulation.

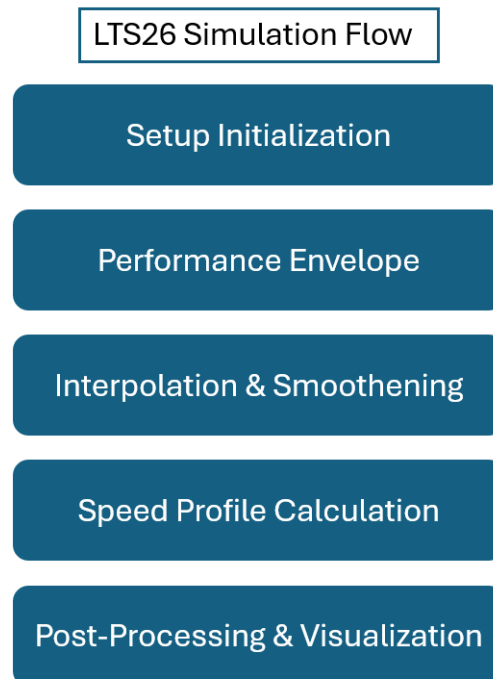


Figure 12 LTS26 Simulation Structure

5.1.3 Course Differentiation

In the Acceleration and Skidpad events, the vehicle operates predominantly under steady-state conditions that is relatively easier to model comparing to the Autocross and Endurance events, which involve continuous and simultaneous variation of both longitudinal and lateral accelerations, representing a considerably more complex dynamic environment

Given that each event type presents unique modelling requirements, the simulation framework in LTS26 was further differentiated beyond the structural overhaul described above. Specifically, the Acceleration and Braking simulations are implemented using a transient model in Simulink, Skidpad to be simulated using a steady-state model, and the Autocross and Endurance simulations are implemented using a quasi-steady-state model in MATLAB scripts. This separation improves both simulation accuracy and numerical stability.

5.2 Transient Model Simulation

5.2.1 Acceleration Simulation

During acceleration, the performance of the race car is governed primarily by two competing limiting factors: tyre traction forces and powertrain driving forces. When the torque delivered by the powertrain exceeds the maximum traction force available at the

driven tyres, wheel spin occurs and tyre grip becomes the binding constraint. Conversely, when tyre grip exceeds the available powertrain output, acceleration becomes power-limited. Accurate prediction of the spatial distribution of these two limiting regimes along the acceleration track is therefore essential for optimising vehicle design parameters and improving event performance.

As the vehicle accelerates, its instantaneous speed increases continuously, which in turn affects both the motor torque output and the aerodynamic forces acting on the vehicle, which vary with the square of velocity. For this reason, vehicle speed is selected as the primary feedback variable within the simulation loop.

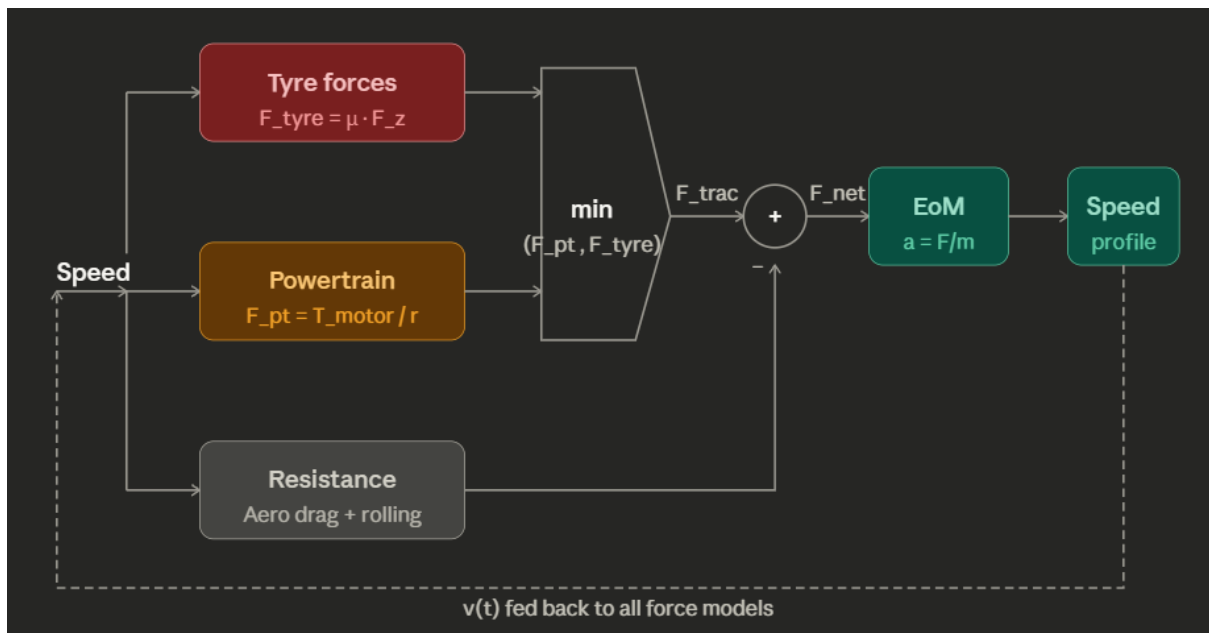


Figure 13 Simplified Design Loop for Acceleration Simulation

5.2.2 Braking Simulation

During braking, vehicle performance is similarly governed by the interaction between tyre forces and the applied braking force. However, since the braking force generated by the hydraulic system can be considered sufficiently large relative to the tyre's capacity to transmit longitudinal force, it is treated as effectively unlimited within this model. Under this assumption, braking performance is simplified to be tyre-limited throughout the event.

As with the acceleration simulation, vehicle speed is adopted as the primary feedback variable, as it directly governs both the aerodynamic loading and the tyre behaviour at each instant during the braking event.

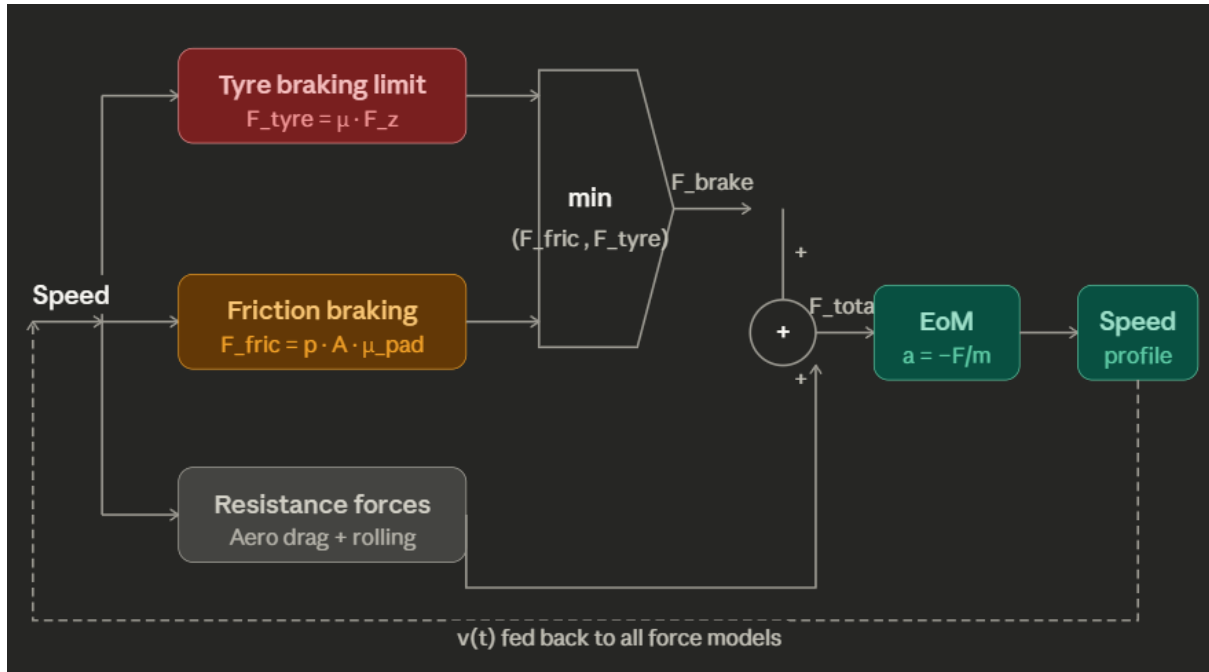


Figure 14 Simplified Design Loop for Braking Simulation

5.3 Quasi-Steady-State Simulation

5.3.1 Vehicle Model

During cornering, the centripetal forces required to sustain the vehicle's curved motion are generated collectively by the four tyres, each contributing forces of varying magnitude and direction depending on the prevailing operating conditions. The tyre force at each corner is a function of slip angle, slip ratio, normal load, inclination angle, and tyre-specific parameters characterised from experimental data.

$$F_{Tire} = f(\text{Slip Angle}, \text{Slip Ratio}, \text{Tire Load}, \text{Inclination Angle}, \text{Tire Parameters},)$$

To determine the resultant tyre forces, several intermediate quantities must be evaluated. Among the most critical is the slip angle, which governs the generation of lateral tyre force and is defined based on the decomposition of the wheel's velocity vector into longitudinal and lateral components relative to the tyre contact patch.

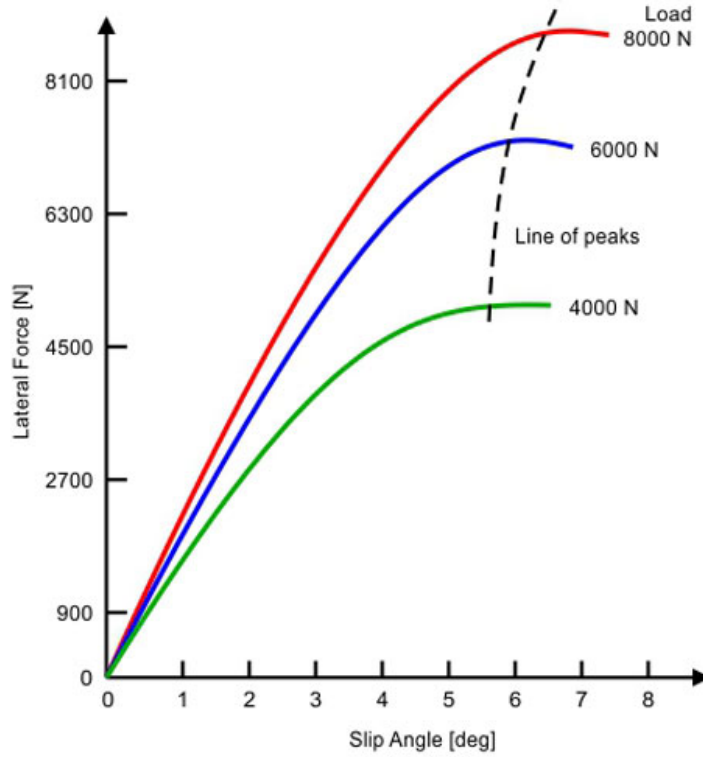


Figure 15 Lateral Force versus Slip Angle (Tyre Slip Angle, 2017)

Slip angle is defined based on the decomposition of the velocity vector into longitudinal and lateral components relative to the tire.

$$\alpha_{FR} = -\delta + \arctan\left(\frac{V_y + \omega_r * a}{V_x + \omega_r * d/2}\right)$$

$$\alpha_{FL} = -\delta + \arctan\left(\frac{V_y + \omega_r * a}{V_x - \omega_r * d/2}\right)$$

$$\alpha_{RR} = \arctan\left(\frac{V_y - \omega_r * b}{V_x + \omega_r * d/2}\right)$$

$$\alpha_{RL} = \arctan\left(\frac{V_y - \omega_r * b}{V_x - \omega_r * d/2}\right)$$

Where, δ is steering angle, ω_r is vehicle's yaw rate, a is distance from vehicle's center of gravity (CG) to the front axle, d is distance from vehicle's CG to the rear axle,

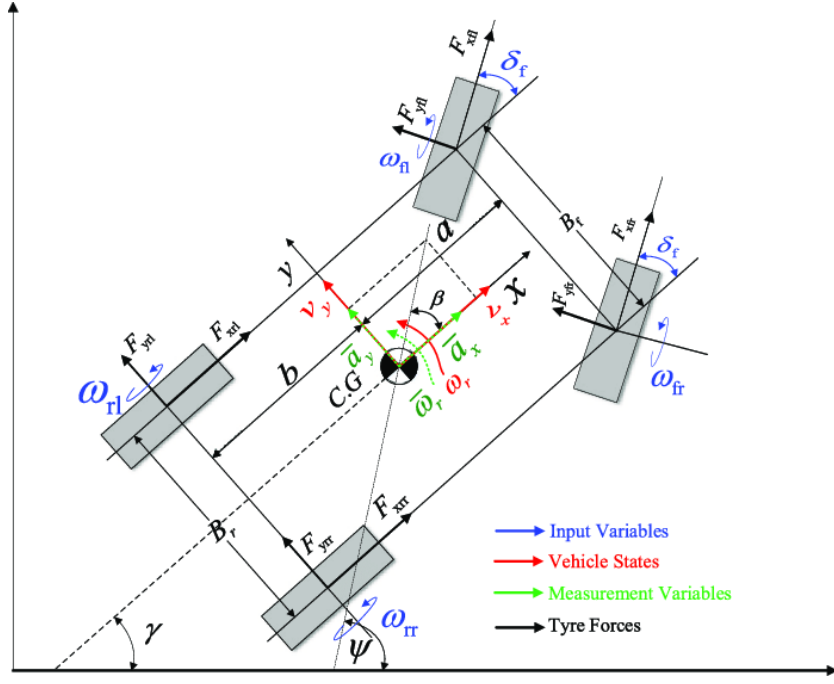


Figure 16 3DOF Vehicle Body Model (Wei, 2016)

Decomposed velocity vector, V_x and V_y are derived from the vehicle's speed and body slip angle β :

$$V_x = V * \cos(\beta)$$

$$V_y = V * \sin(\beta)$$

A second critical input to the tyre force model is the normal load at each corner, which arises from four contributing sources.

$$F_z = \frac{Mass * 9.81}{4}$$

$$F_{Aero} = \frac{1}{4} * (0.5 * Density * CL * Area * V^2)$$

$$F_{Lat_LT} = F_z * \frac{A_y * Cg_h}{9.81 * d}$$

$$F_{Long_LT} = F_z * \frac{A_x * Cg_h}{9.81 * wheelbase}$$

Where, $Mass$ is vehicle's total mass, $Density$ is the density of fluid (in this case, air), CL is the Coefficient of Lift, $Area$ is vehicle's frontal area, V is vehicle's speed value, A_y is vehicle's Lateral Acceleration, A_x is vehicle's Longitudinal Acceleration, Cg_h is the height of vehicle's CG, and d is vehicle's track length. Here we assume that the vehicle has equally distributed static weight and aerodynamic downforce across four wheels.

Assuming the vehicle is negotiating a left-hand turn while accelerating, resulting normal load at the rear-right tyre is accordingly given by:

$$F_{zrr} = F_z + F_{Aero} + F_{Lat_LT} + F_{Long_LT}$$

Sign conventions are adjusted accordingly for the remaining wheel positions. Tyre inclination angle is assumed to be zero in the current model, given its relatively minor contribution to tyre force generation at this level of modelling fidelity.

Tyre characteristics are defined using pre-fitted parameters derived from experimental measurements and are treated as fixed known inputs to the optimisation problem.

Upon evaluating the longitudinal and lateral forces at each of the four tyres, the complete four-wheel vehicle dynamic geometry is referenced to compute the resultant longitudinal force, lateral force, and yaw moment with respect to the fixed vehicle coordinate frame.

$$F_y = F_{yFR} * \cos(\delta) + F_{yFL} * \cos(\delta) + F_{xFR} * \sin(\delta) + F_{xFL} * \sin(\delta) + F_{yRL} + F_{yRR}$$

$$F_x = F_{xFR} * \cos(\delta) + F_{xFL} * \cos(\delta) - F_{yFR} * \sin(\delta) - F_{yFL} * \sin(\delta) + F_{xRL} + F_{xRR}$$

$$\begin{aligned} M_z = & a * (F_{xFR} * \sin(\delta) + F_{xFL} * \sin(\delta) + F_{yFR} * \cos(\delta) + F_{yFL} * \cos(\delta)) \\ & - b * (F_{yRL} + F_{yRR}) \\ & + \frac{d}{2} * (F_{xFR} * \cos(\delta) - F_{xFL} * \cos(\delta)) \\ & + \frac{d}{2} * (F_{xRR} - F_{xRL}) \\ & + \frac{d}{2} * (F_{yFL} * \sin(\delta) - F_{yFR} * \sin(\delta)) \end{aligned}$$

The vehicle accelerations with respect to the path-tangential coordinate frame are then given by:

$$a_x = \frac{1}{m} * (F_y \sin(\beta) + F_x \cos(\beta) - Drag)$$

$$a_y = \frac{1}{m} * (F_y \cos(\beta) - F_x \sin(\beta))$$

At this stage, several variables remain undetermined, specifically the steering angle δ , body slip angle β , yaw rate ω_r , and the slip ratios at each tyre. These quantities are treated as decision variables within the optimisation problem. To ensure internal consistency between the assumed and computed accelerations, residual equality constraints are introduced, enforcing that the acceleration values used in the load transfer calculations are consistent with those produced by the vehicle model upon solution:

$$A_{xRes} = A_x - A_{x_in}$$

$$A_{yRes} = A_y - A_{y_in}$$

5.3.2 Dynamic Model

5.3.2.1 Steady-State Model

As the goal being constructing a G-G diagram that captures the vehicle's maximum potentials, combinations of longitudinal and lateral accelerations values need to be output from simulation to cover all possible scenarios during dynamic events. This will be achieved by performing numerical optimization using the Vehicle Model formulated above with specific set of decision variables.

A fundamental consideration in the design of the steady-state model is that numerical optimisation, by formulation, permits only a single objective variable to be minimised or maximised per solver call, and the remaining unknowns are determined as a consequence of satisfying the imposed constraints. The core algorithmic challenge is therefore to design a structured approach that ensures both A_x and A_y are simultaneously driven towards their maximum achievable values, despite the solver being capable of directly optimising only one at a time.

The first design iteration addressed this by decomposing the problem into a sequential series of optimisation calls. Since tyre grip is most efficiently utilised when directed along a single axis, the maximum and minimum achievable longitudinal accelerations are first determined in isolation by setting lateral acceleration to zero. These two optimisation problems define the feasible range of A_x values. The programme subsequently generates a uniform mesh of A_x values between these bounds, and for each discrete value imposes A_x as a fixed constraint while maximising A_y as the objective. This yields a set of optimised (A_x, A_y) pairs that collectively trace the G-G boundary.

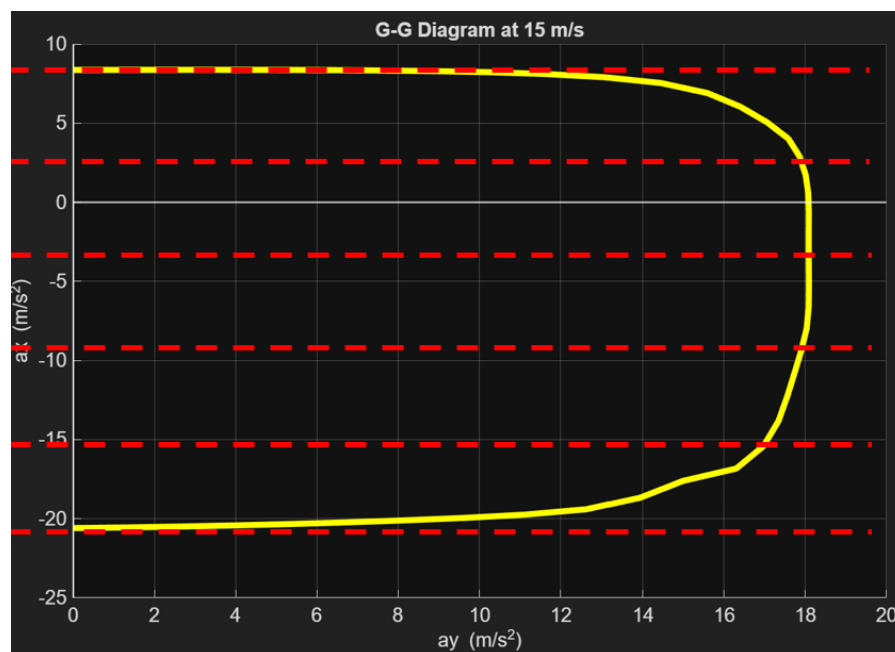


Figure 17 G-G Diagram First Prototype Illustration

While conceptually straightforward, this formulation required a minimum of three separate nonlinear optimisation problem configurations, increasing implementation complexity. More critically, the rectangular mesh imposed on A_x did not align naturally with the circular geometry of the G-G boundary, resulting in query points near the pure braking and pure acceleration extremes being sparsely distributed relative to the combined acceleration region. This mismatch caused a high rate of optimisation failure at the boundary regions where the feasible set is most constrained, and produced uneven resolution across the G-G diagram.

To address these shortcomings, a revised formulation was adopted in which the optimisation objective is no longer defined in terms of either longitudinal or lateral acceleration independently, but instead in terms of a scalar grip factor P , defined as:

$$|P| = \sqrt{Ax^2 + Ay^2}$$

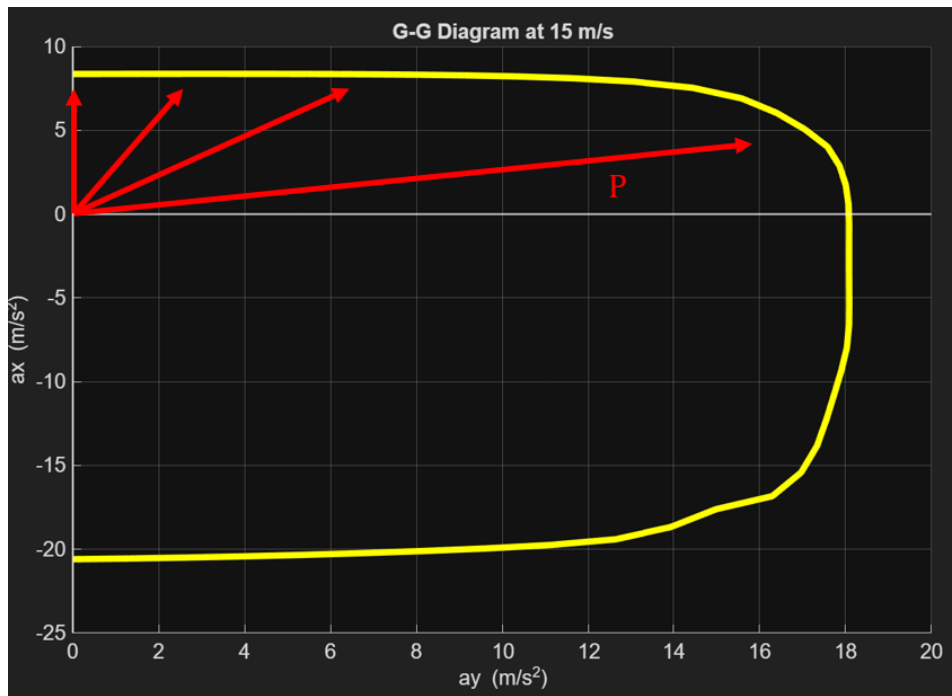


Figure 18 G-G Diagram Second Prototype Illustration

The programme defines an angular sweep variable θ uniformly distributed over $[-\pi, \pi]$. This polar formulation naturally distributes query points evenly along the G-G boundary, reduces the problem to a single unified optimisation structure, and eliminates the need to pre-determine the A_x bounds. The angular mesh increment was selected by balancing diagram resolution against computational cost, and initial guesses at each instance are warm-started from the preceding solution to exploit the expected smoothness of the boundary and improve convergence reliability.

A full description of the numerical optimization setup can be found in Appendix A.6.

5.3.2.2 Quasi-Steady State Model

To simulate the complete performance of a race car competing in events such as Autocross and Endurance, the Steady-State Model is repeated systematically across the full operational speed range of the vehicle. This procedure constructs a G-G-V diagram that characterises the vehicle's dynamic limits across all operating conditions.

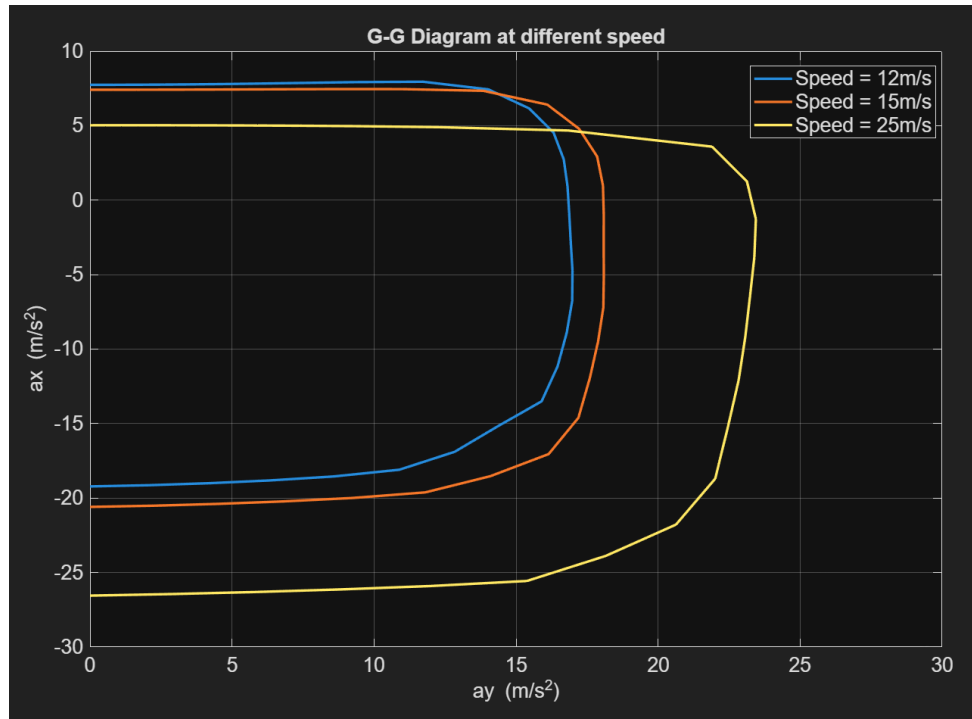


Figure 19 Simulated G-G Diagram at different speeds

The G-G diagram computations were performed at speeds ranging from 10 m/s up to the vehicle's theoretical maximum speed, with a uniform increment of 1 m/s. Speeds below 10 m/s were excluded from direct computation since it's uncommon to occur in FSAE competition, and the vehicle has significant different behaviours at such low speeds. To account for this excluded regime, linear extrapolation was applied to the computed

results immediately above the 10 m/s boundary to estimate the vehicle's performance below this speed.

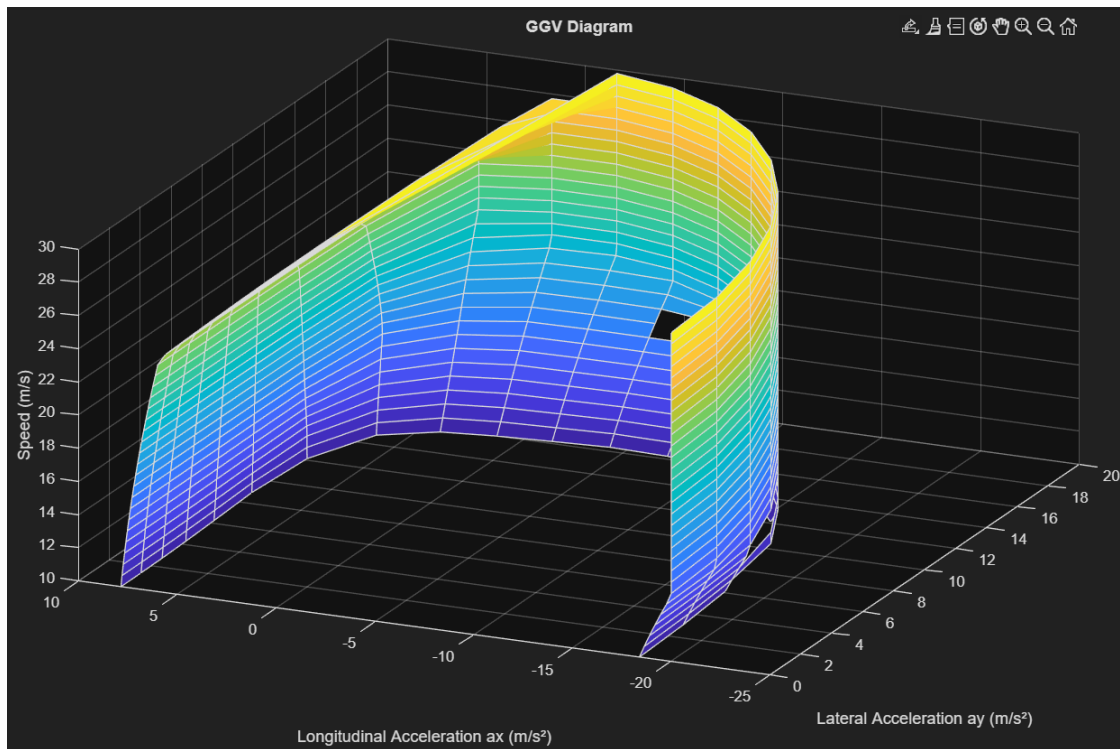


Figure 20 Interpolation of G-G-V Diagram

5.3.2.3 Dynamic Event Simulation

With the vehicle's performance limits established through the G-G-V diagram, it becomes possible to predict the vehicle's dynamic capabilities on an actual competition track, as encountered during the Autocross and Endurance events. The simulation proceeds in two stages: first, a Boundary Speed Profile (BSP) is generated to represent the maximum achievable cornering speed at each point along the track; second, the maximum achievable speed accounting for the constraints imposed by acceleration and braking is computed, yielding the Limit Speed Profile (LSP).

To construct the BSP, the programme determines the maximum achievable lateral acceleration A_y at each speed value from the interpolated G-G-V diagram, representing the vehicle's peak cornering capability at that speed. The minimum achievable turn radius for each speed-acceleration combination is then computed as:

$$R = \frac{V^2}{A_y}$$

Equation 1 Turn Radius

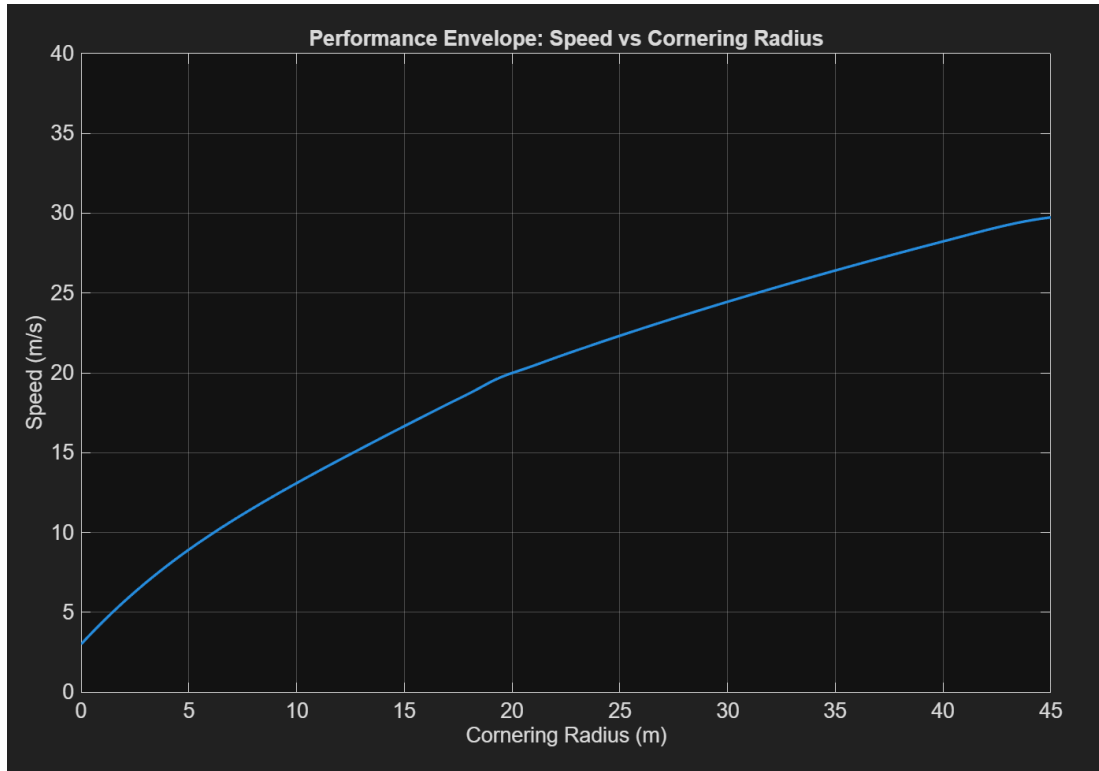


Figure 21 Cornering Speed against Radius

The discrete results are subsequently interpolated to generalise the relationship across all cornering scenarios. The programme maps the cornering speed to the corresponding turn radius for each track segment with a turn radius below 50 metres; for segments with a turn radius exceeding this threshold, the vehicle's maximum speed is assigned directly. The BSP is thereby obtained by mapping the cornering speed values with track curvatures.

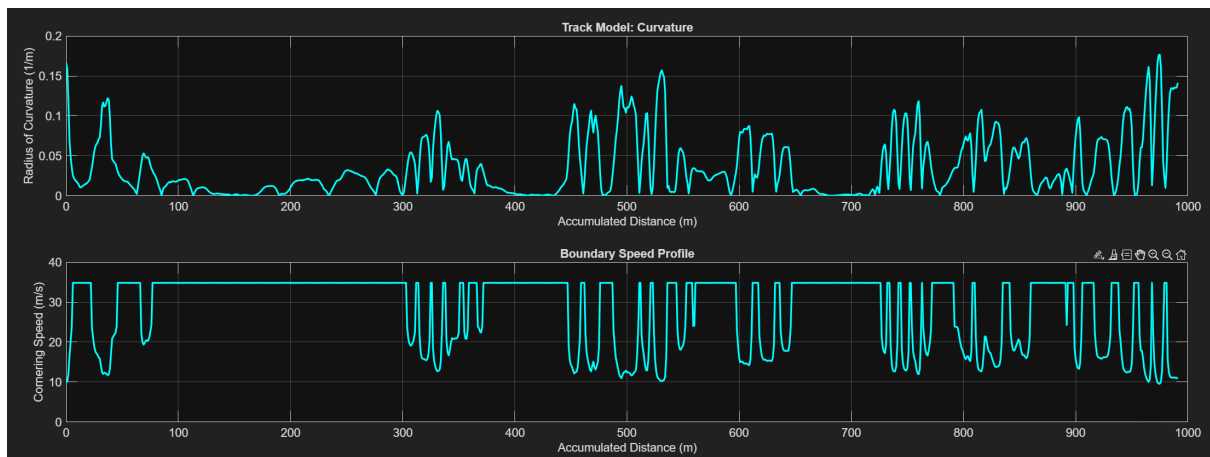


Figure 22 Track Model and Boundary Speed Profile

The simulation then proceeds to determine the vehicle's maximum achievable longitudinal acceleration at each track position, in order to predict the highest attainable speed at the subsequent location following acceleration. Given the speed and turn radius prescribed by the BSP, and with the lateral acceleration accordingly known, the corresponding longitudinal acceleration A_x is extracted from the G-G-V diagram. Since

the G-G-V diagram encompasses both acceleration and braking regimes, a given combination of speed and lateral acceleration yields two distinct values of A_x — one positive, corresponding to acceleration, and one negative, corresponding to braking. The G-G-V diagram is therefore partitioned into two separate regions to enable independent extraction of acceleration and braking data.

Two sequential iteration loops are implemented to update the speed values derived from the Boundary Speed Profile. The first loop evaluates the maximum achievable longitudinal acceleration at each instance using the acceleration Performance Envelope, while the second loop applies the corresponding braking deceleration. The complete Limit Speed Profile is thereby obtained as the final output of the dynamic event simulation.

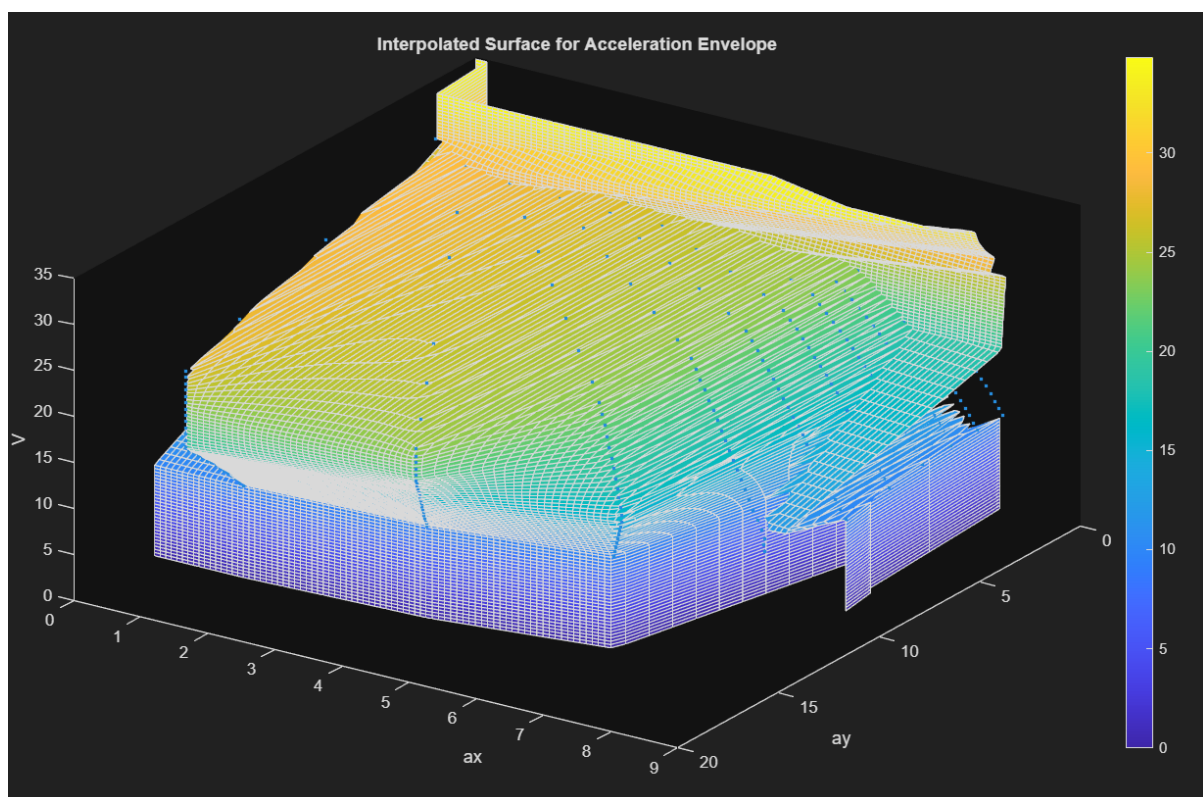


Figure 23 Performance Envelope for Acceleration

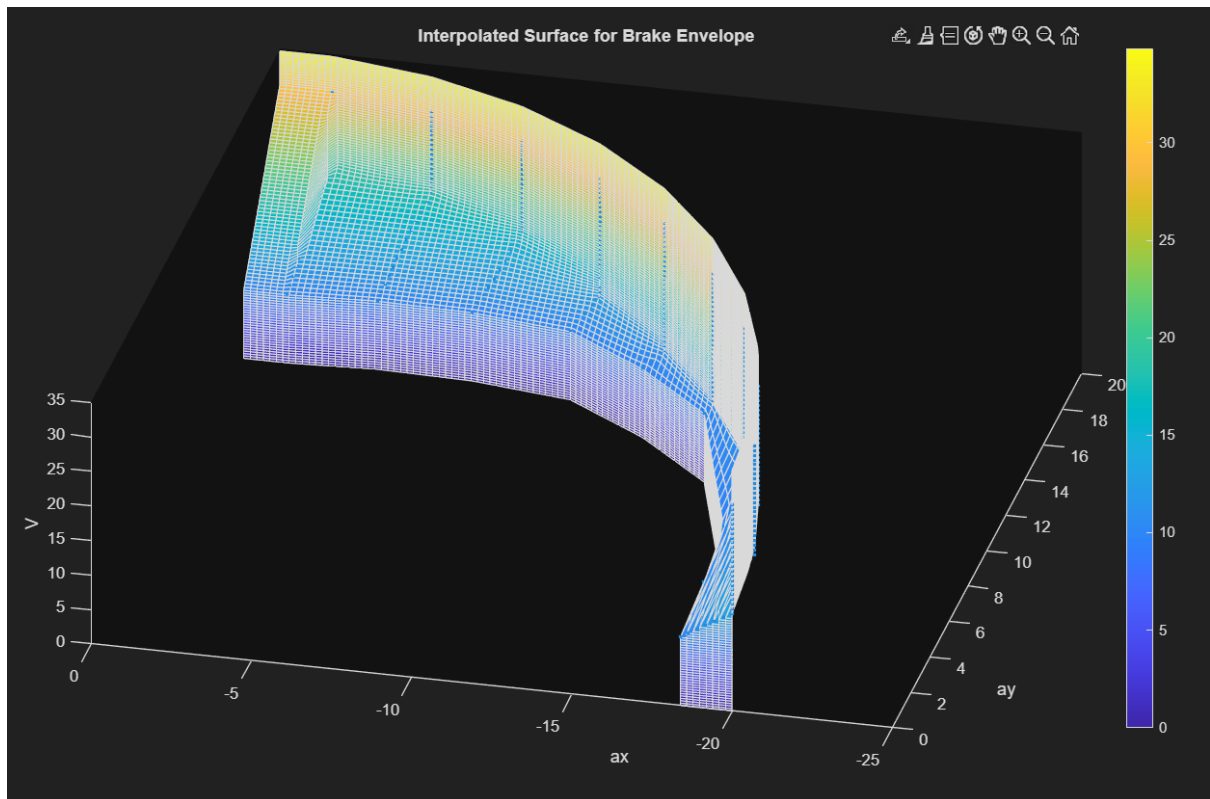


Figure 24 Performance Envelope for Braking



Figure 25 Limit Speed Profile

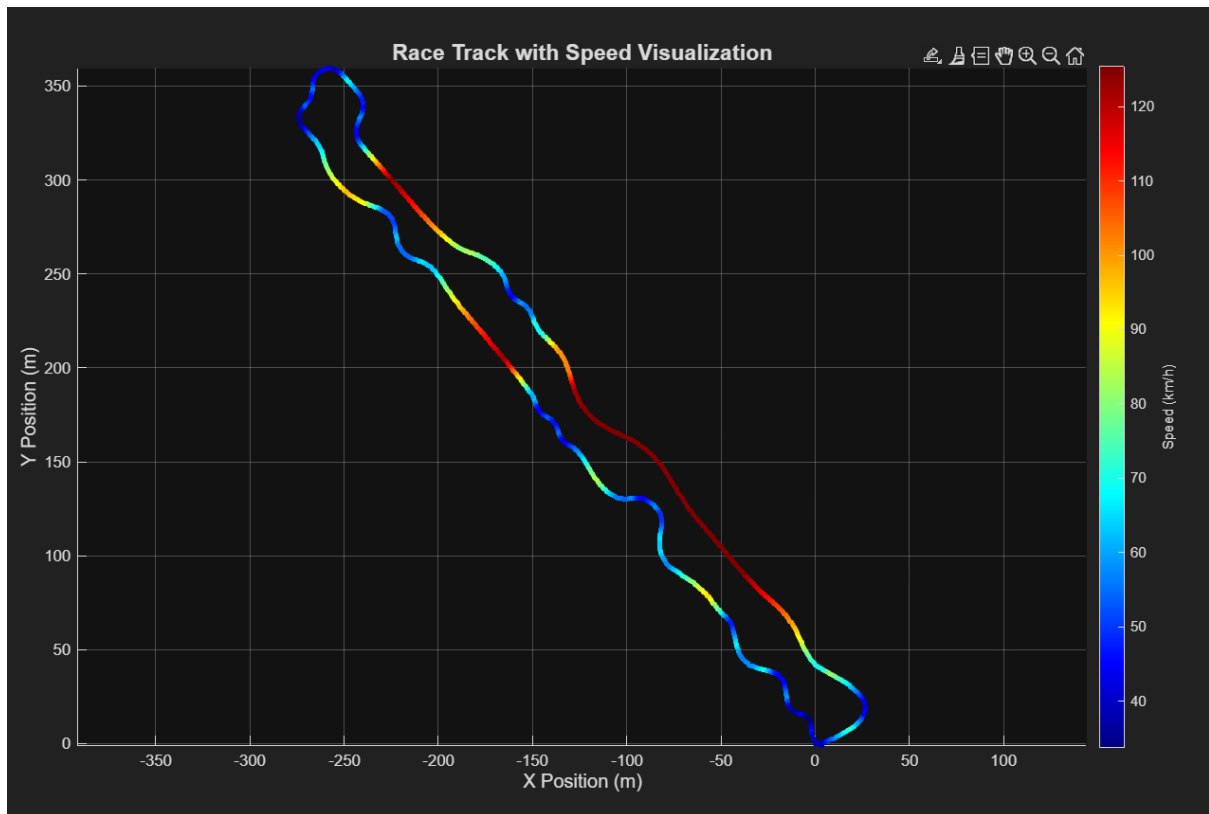


Figure 26 Race Track with Speed Visualization

In addition to speed values, more vehicle parameters can be simulated to provide further insights to the team.

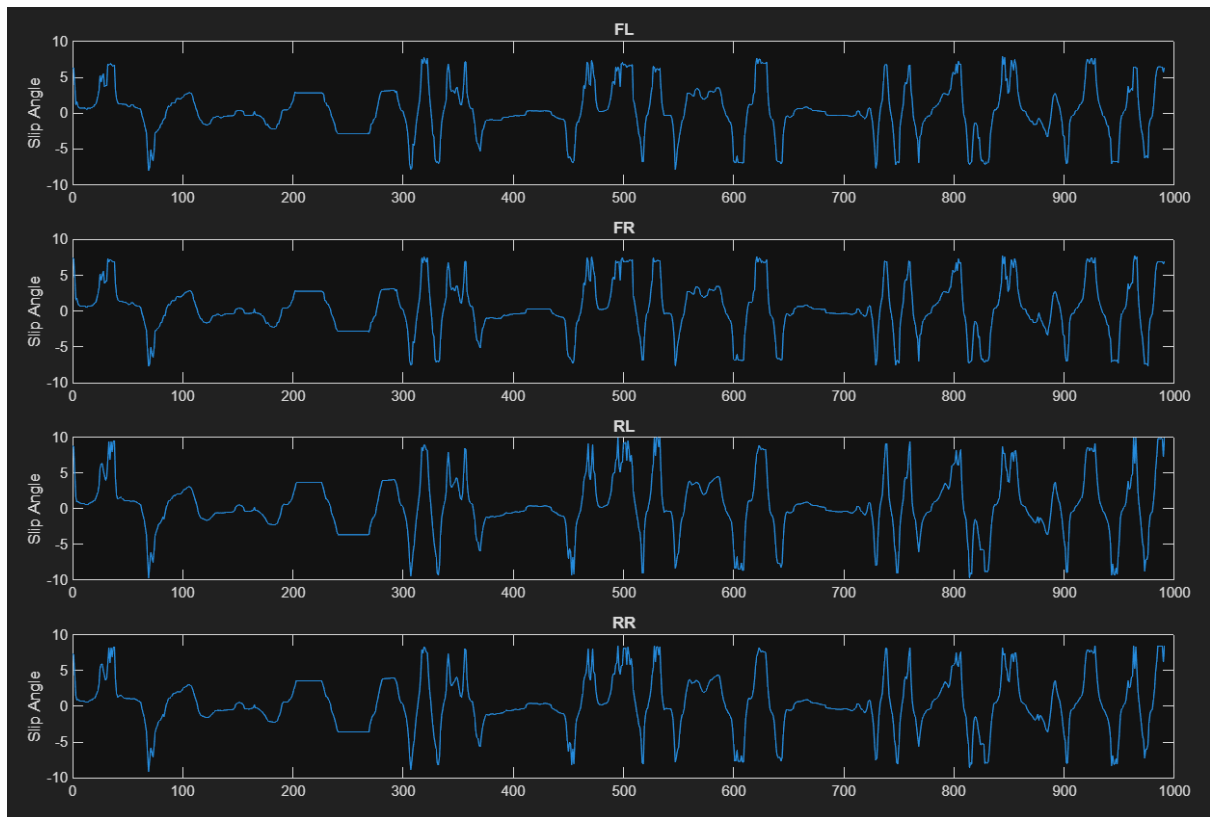


Figure 27 Simulation of Slip Angle Variations

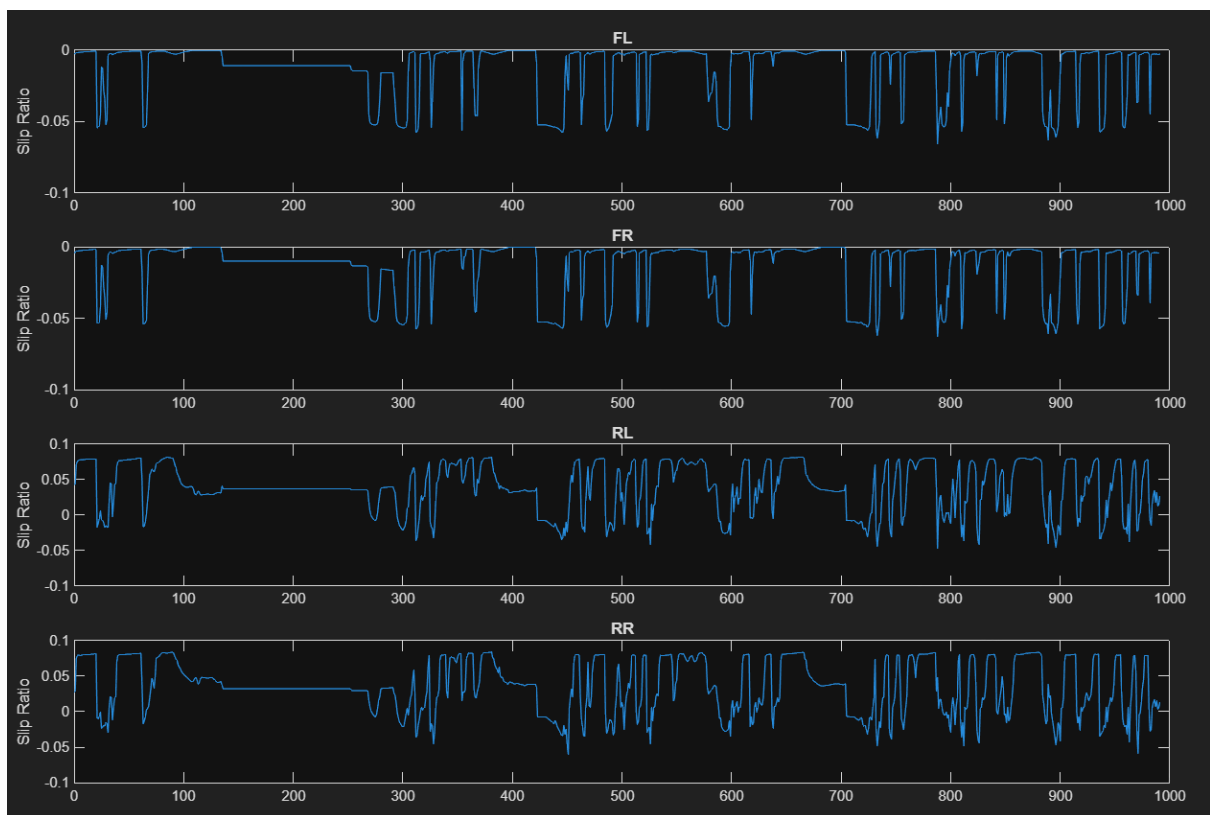


Figure 28 Simulation of Slip Ratio Variations

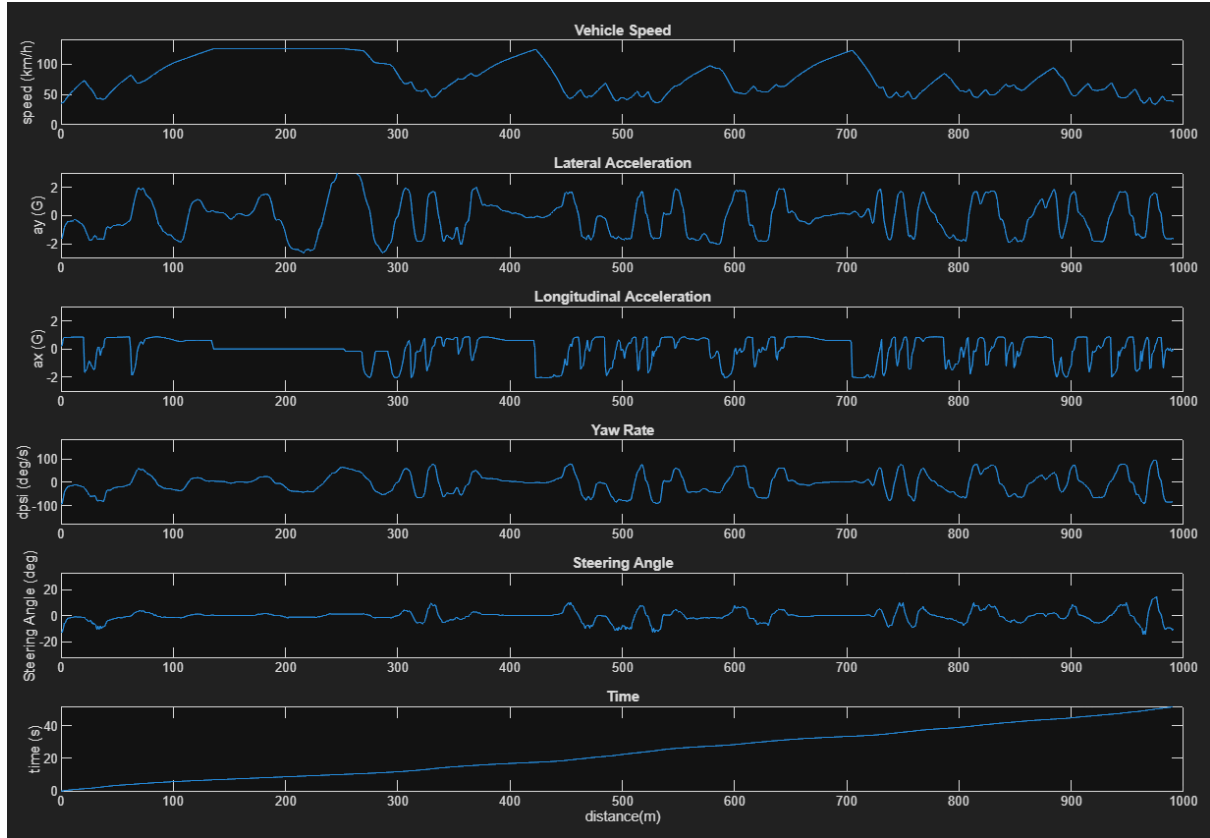


Figure 29 Simulation of common vehicle parameters

5.3.3 Subsystem Simulation Implementation

5.3.3.1 Aerodynamics

Aerodynamic properties are provided by the Aerodynamics department based on Computational Fluid Dynamics (CFD) simulations conducted using a third-party solver. Simulations are performed across a range of flow directions and vehicle geometries to account for both straight-line and cornering operating conditions, such that the aerodynamic coefficients vary accordingly between these regimes. While it is desirable to further differentiate the aerodynamic properties as a function of cornering radius, any such differentiation must be implemented in a manner that preserves the linearity and differentiability of the optimisation objective function. To satisfy this requirement, a sigmoid function is employed in place of conventional if-else conditional logic.

$$\sigma = \frac{1}{1 + e^{-k(\sqrt{\delta_{in}^2 + \varepsilon} - \frac{2\pi}{180})}}$$

$$CL = CL_s(1 - \sigma) + CL_c * \sigma$$

$$CD = CD_s(1 - \sigma) + CD_c * \sigma$$

$$ab = ab_s(1 - \sigma) + ab_c * \sigma$$

5.3.3.2 Steering System

Ackermann steering geometry is widely adopted to differentiate the steering angles of the inner and outer front wheels during cornering, thereby improving directional stability and reducing tyre scrub. The simulation of Ackermann geometry effects on vehicle dynamic performance is therefore necessary for accurate modelling.

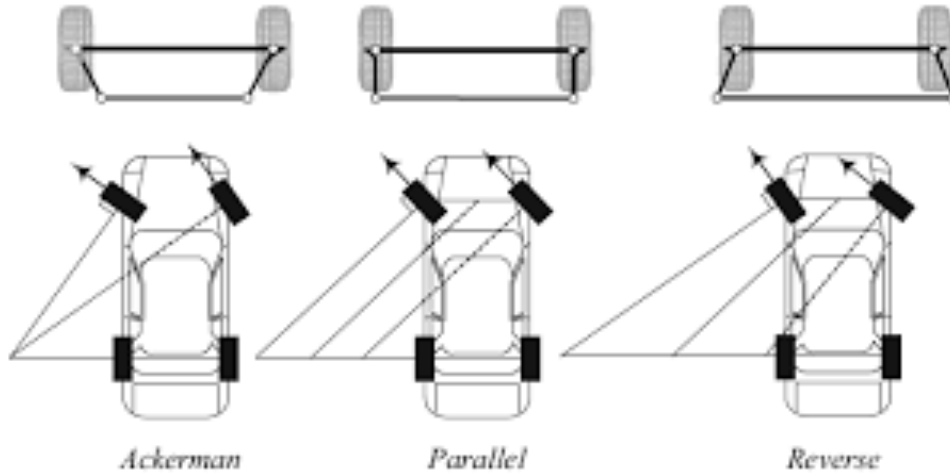


Figure 30 Ackermann Geometry Comparison (Deshmukh, 2021)

The Ackermann geometry is characterised using Optimum Kinematic software, which generates the relationship between the left and right steering angles and the percentage of maximum steering input as a function of the expected cornering radius. Polynomial curve fitting is subsequently applied to this data to establish a continuous functional relationship between the inner and outer wheel steering angles and the instantaneous cornering radius. Incorporating this relationship into the slip angle formulation modifies the front axle slip angle expressions as follows:

$$\alpha_{FR} = -\delta_{Inner} + \arctan\left(\frac{V_y + \omega_r * a}{V_x + \omega_r * d/2}\right)$$

$$\alpha_{FL} = -\delta_{Outer} + \arctan\left(\frac{V_y + \omega_r * a}{V_x - \omega_r * d/2}\right)$$

$$\alpha_{RR} = \arctan\left(\frac{V_y - \omega_r * a}{V_x + \omega_r * d/2}\right)$$

$$\alpha_{RL} = \arctan\left(\frac{V_y - \omega_r * a}{V_x - \omega_r * d/2}\right)$$

This now enables the comparison between different Ackermann geometry as shown below.

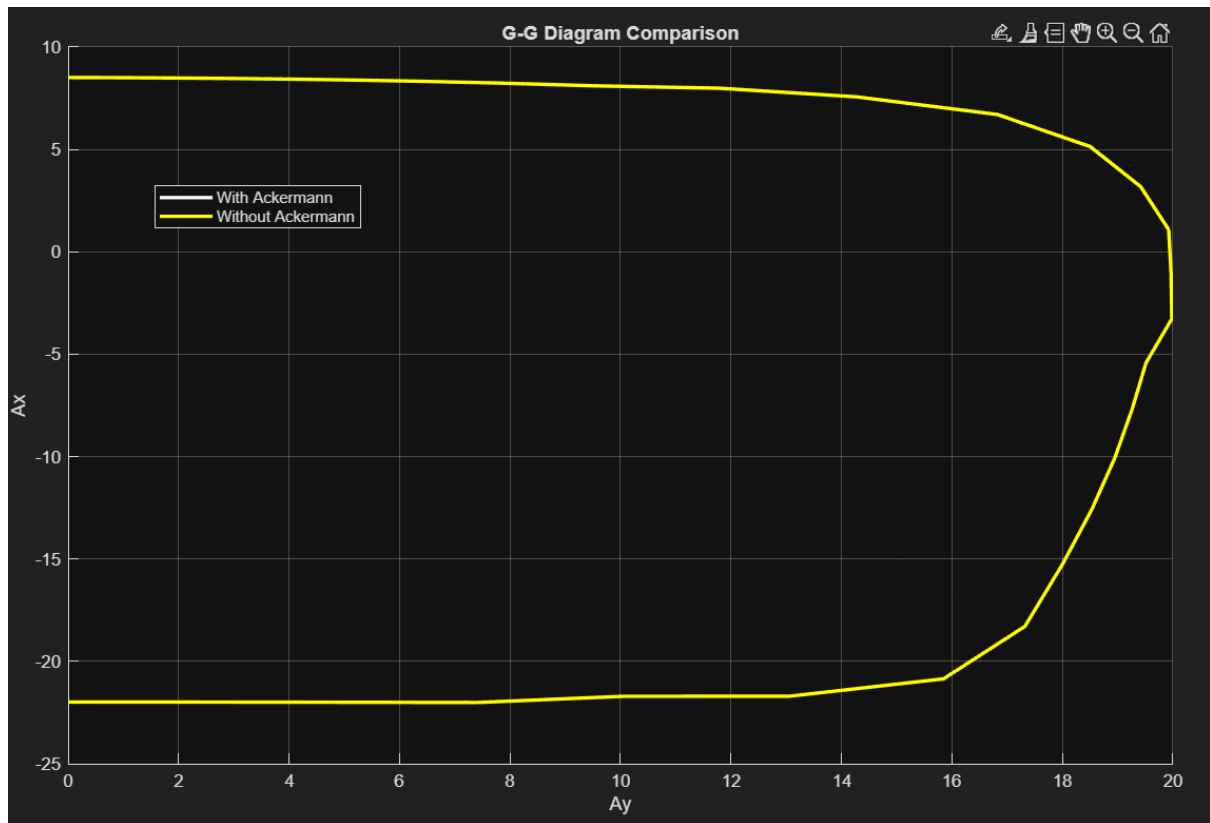


Figure 31 G-G Diagram Comparison with Ackermann

Simulation results comparing the vehicle performance with and without the Ackermann steering correction were evaluated. Upon overlaying the two outputs, the resulting G-G diagrams were found to be almost entirely coincident, with differences manifesting only at the level of decimal precision.

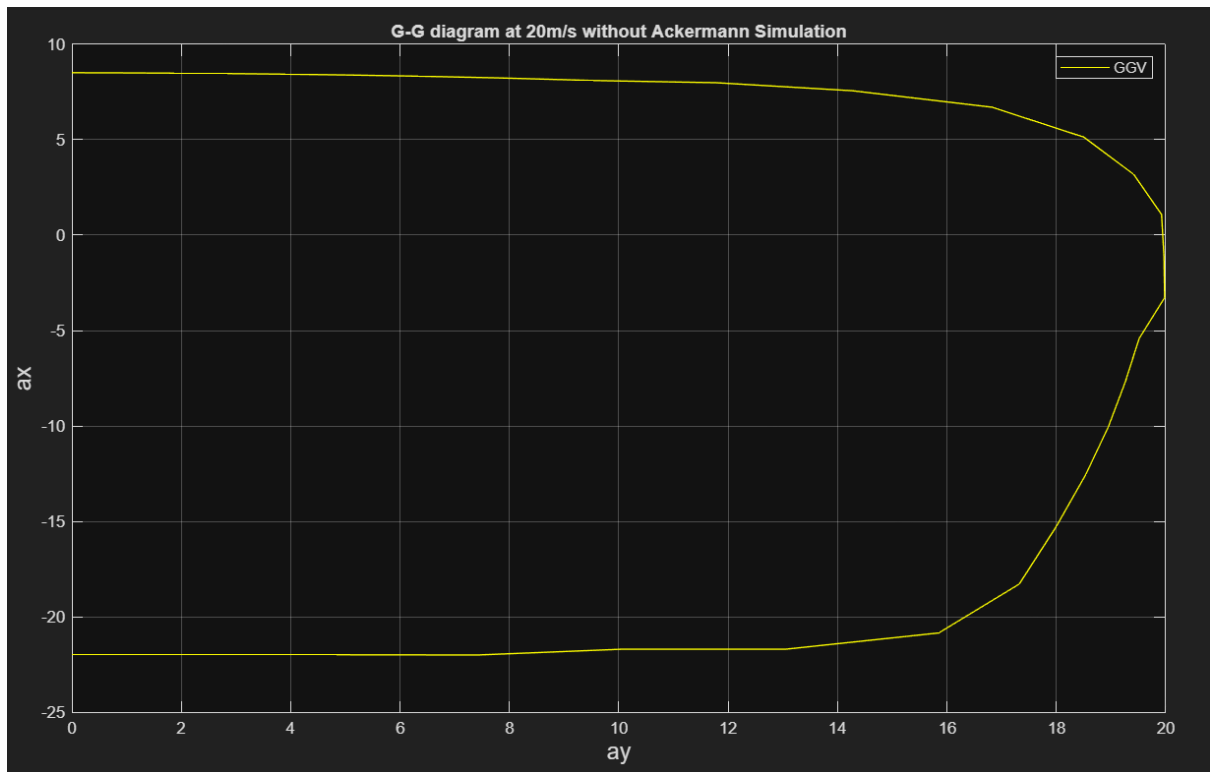


Figure 32 G-G Diagram without Ackermann Simulation

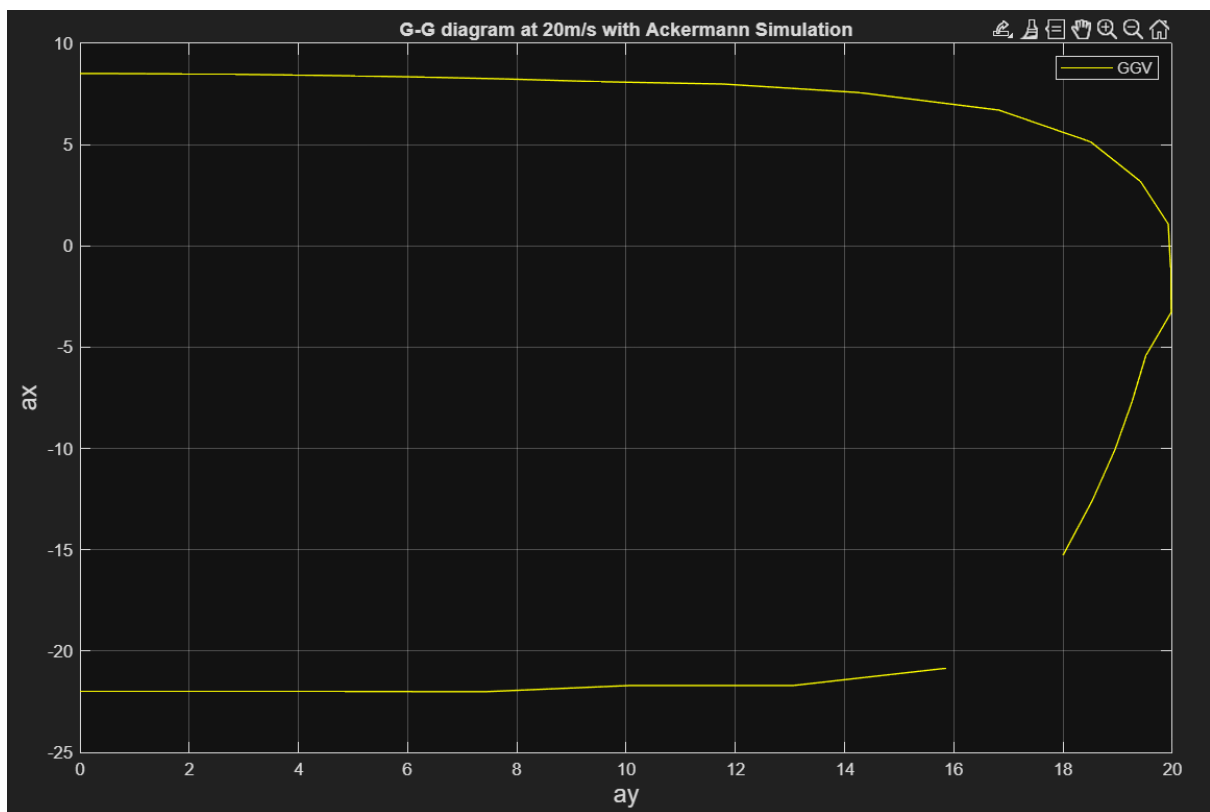


Figure 33 G-G Diagram with Ackermann Simulation

It was further observed that the introduction of the Ackermann steering feature introduces additional numerical complexity into the optimisation problem, arising from

the higher-order polynomial curve fitting function used to represent the steering geometry.

Given that the performance benefit of the Ackermann steering feature is marginal while its adverse effect on numerical stability is significant, it was decided that this feature would be retained exclusively for steady-state G-G diagram simulation and deliberately excluded from the quasi-steady-state lap simulation.

5.3.3.3 Brake System

During braking, longitudinal load transfer shifts the normal load distribution towards the front axle. To fully exploit the available tyre grip under these conditions, it is advantageous to distribute the braking force asymmetrically between the front and rear axles, commonly referred to as brake bias. Determining the optimal brake bias for the NUS FSAE vehicle under its specific operating conditions is therefore of direct engineering interest, and a dedicated simulation capability was developed to address this requirement.

The brake bias constraint is implemented by imposing an additional equality constraint within the numerical optimisation framework, requiring that the ratio of front to total braking force is equal to the prescribed brake bias value during braking.

$$Brake\ Bias = \frac{F_{xFL}}{F_{xFL} + F_{xRL}}$$

By adding this constraint, it is observed from the G-G diagram comparison that the introduction of a fixed brake bias reduces the peak longitudinal deceleration relative to the unbiased baseline. As lateral acceleration increases, aerodynamic and inertial load transfer shifts the tyre normal forces away from the balance point assumed by the fixed bias, causing one axle to saturate before the other reaches its grip limit. The combined braking capacity is therefore lower than the theoretical maximum achievable when both axles are allowed to operate simultaneously at their respective traction limits.

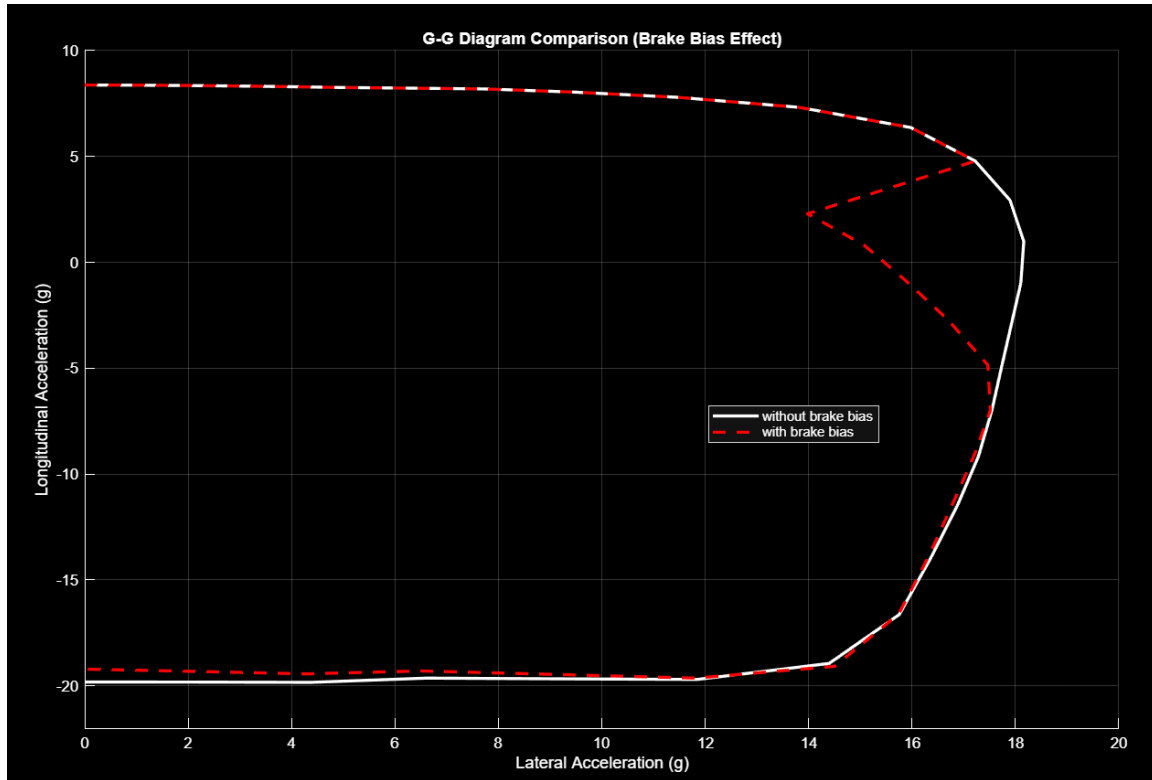


Figure 34 G-G Diagram Comparison with Brake Bias

More critically, the fixed brake bias introduces a discrete regime switch into the simulation model, causing the objective landscape presented to the optimiser to contain a discontinuity at which the gradient of the braking force with respect to the decision variables is undefined. This manifests visibly in the G-G diagram as the inward concavity observed in the high-lateral-acceleration region of the constrained curve, where the optimiser loses the ability to track the outer performance envelope smoothly. The underlying cause of this instability and its impact on the fidelity of the simulation results remains an open issue.

Consequently, the brake bias constraint is deliberately excluded from the quasi-steady-state (QSS) lap simulation in order to preserve its numerical stability and the physical continuity of the resulting speed and force profiles.

6 Testing and Tuning

6.1 Straight Line Simulation Validation

The acceleration simulation was applied to a 75-metre straight-line track representative of the Formula SAE Acceleration event. The resulting speed and force profiles demonstrate behaviour consistent with the expected physical characteristics of the vehicle under full-throttle conditions.

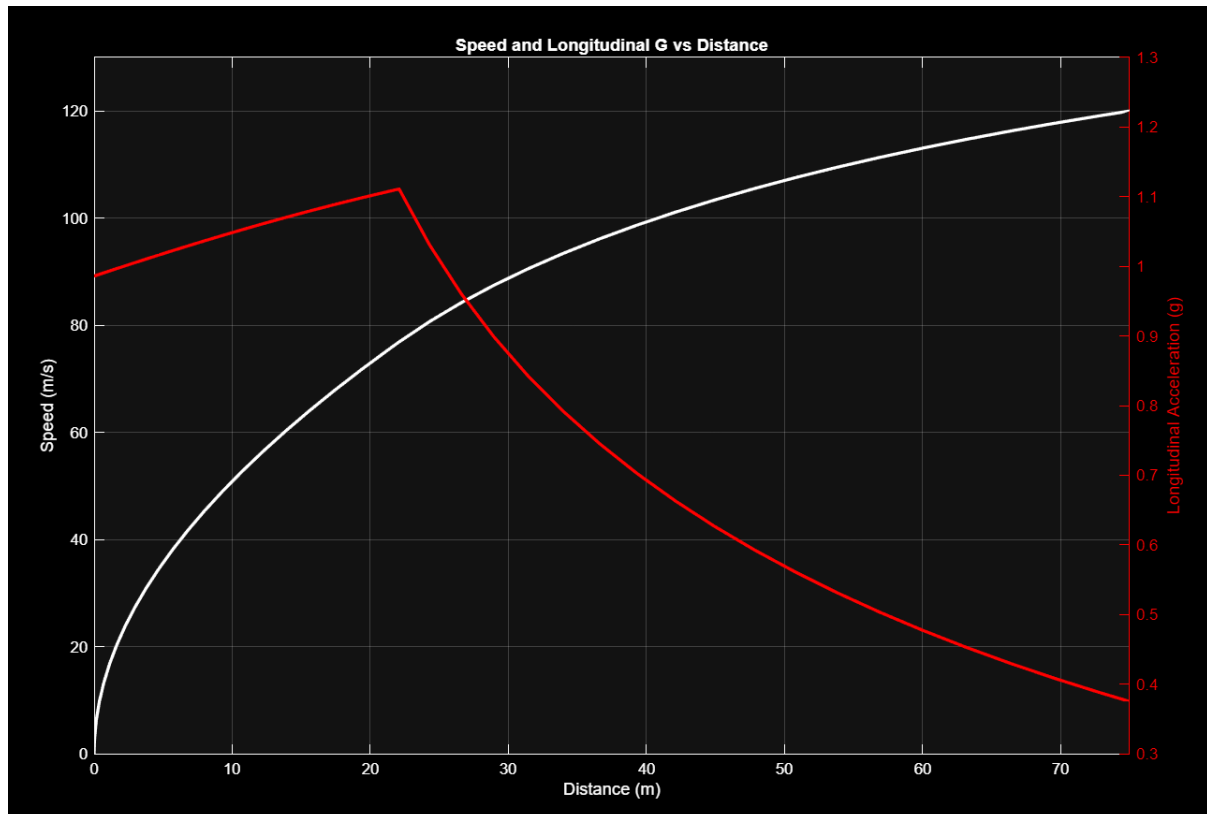


Figure 35 Simulated Speed and Longitudinal G in Acceleration Event

The sensitivity of the simulation to vehicle setup parameters was further examined by varying the Final Drive Ratio (FDR). The results confirm that the simulation responds appropriately to changes in this parameter, producing distinct speed profiles that reflect the trade-off between peak torque delivery and top speed associated with different gear

ratios.

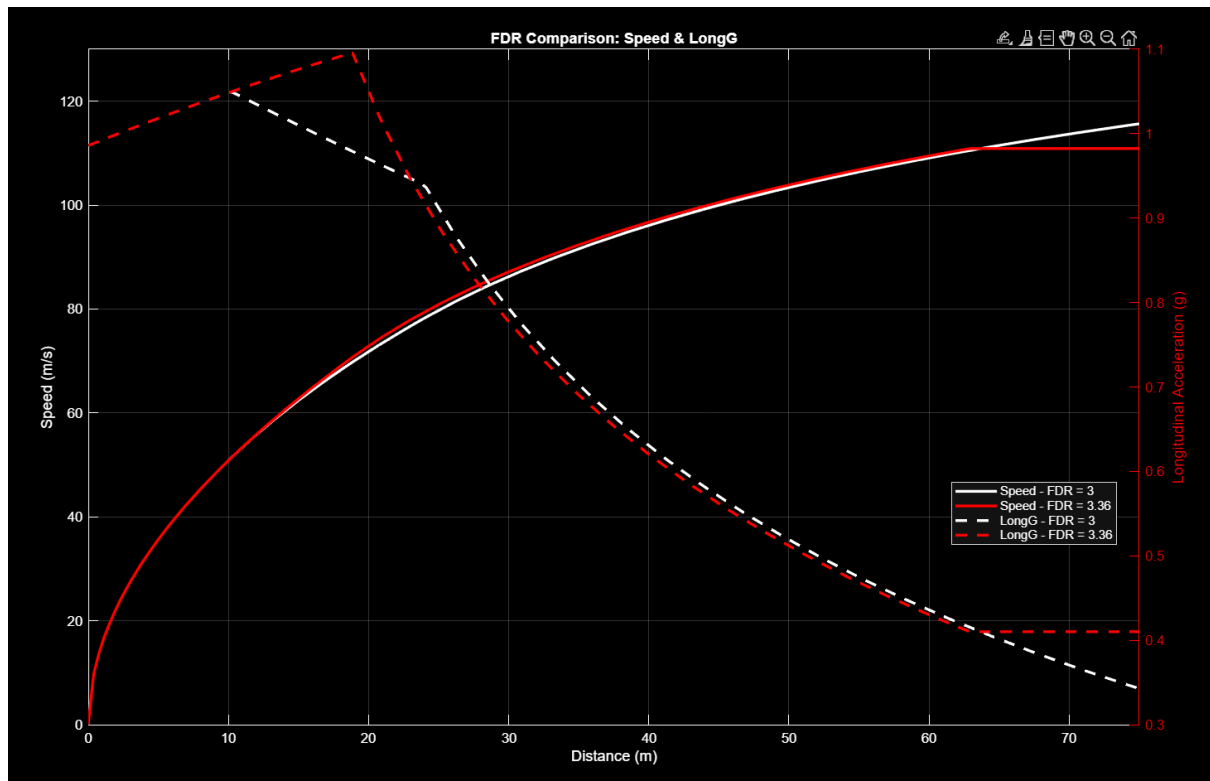


Figure 36 Comparison of FDR in Acceleration Event

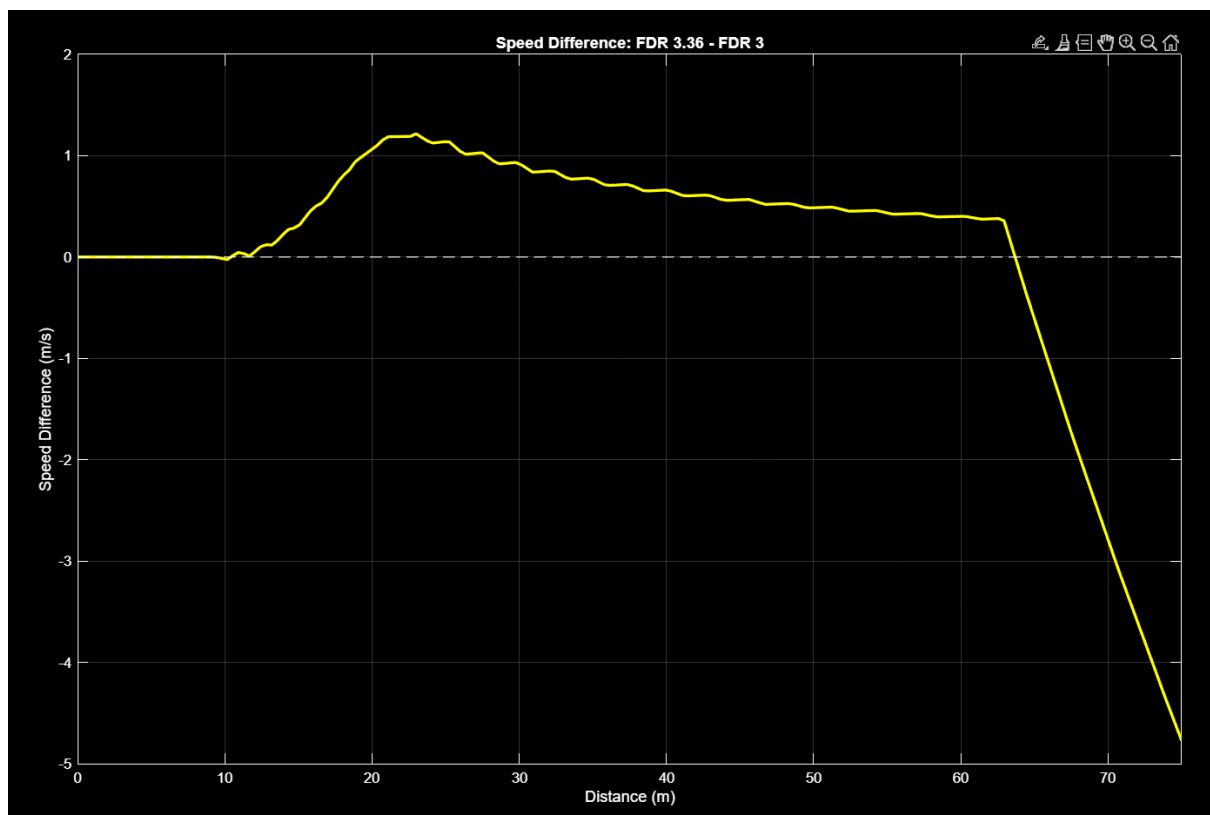


Figure 37 Speed Difference over Distance for different FDR

To validate the accuracy of the model, the simulated speed profile was compared against telemetry data recorded during the Formula SAE Michigan 2025 Acceleration event. Although the recorded telemetry exhibits localised fluctuations attributable to sensor noise, the overall trend and critical performance features are in close agreement with the simulation output.

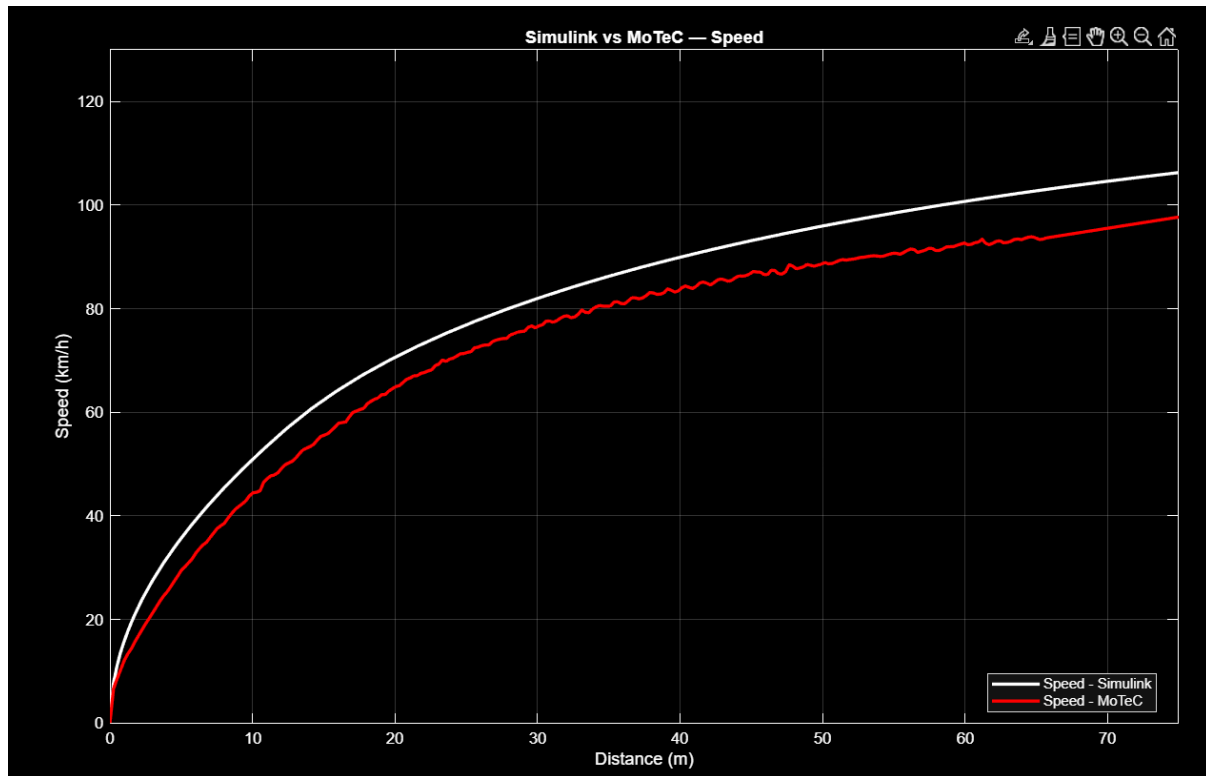


Figure 38 Comparison of Simulated Speed and Vehicle Data

Comparison of the simulated and measured speed profiles reveals a localised discrepancy concentrated in the early phase of acceleration. As shown in the speed error plot, the divergence is not sustained throughout the run but is instead confined to the initial distance window. This behaviour is attributed to wheel spin observed in the physical vehicle during launch. The simulation assumes ideal tyre–road contact with no slip beyond the tyre model's steady-state limits, and therefore does not replicate the transient speed loss associated with a wheelspin event.

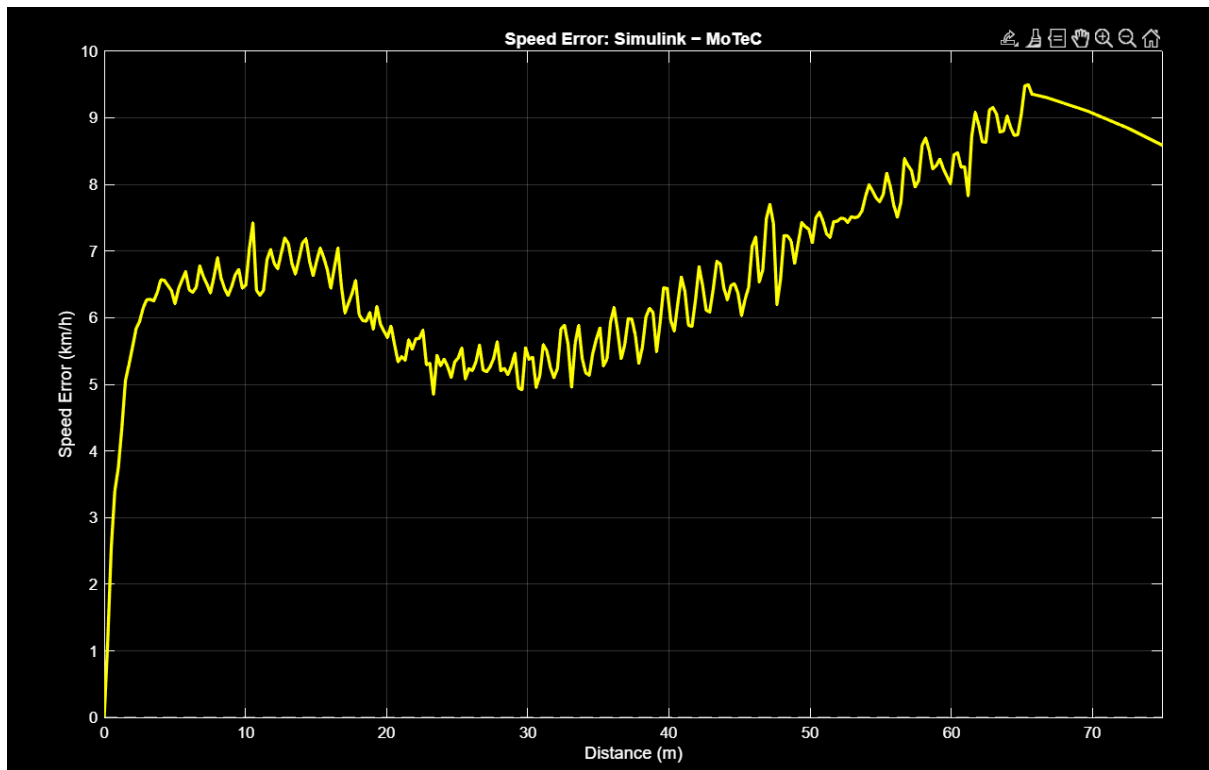


Figure 39 Speed Prediction Error over Distance

6.2 Corner Simulation Validation

6.2.1 Numerical Optimization Tuning

At each point of interest within the simulation, a separate numerical optimisation is performed to determine the optimal set of decision variables. Since a generalised configuration of numerical constraints and initial guesses is applied uniformly across all instances, certain optimisation problems may fail to converge due to local infeasibility or the highly nonlinear nature of the objective landscape in certain regions of the operating domain. To prevent isolated failures from interrupting the overall simulation process, the programme skip any failed instances automatically and continue execution. Upon completion of the simulation, an optimisation success rate is computed as the ratio of successfully converged instances to the total number of optimisation attempts, providing a quantitative metric for assessing the overall credibility and reliability of each simulation run.

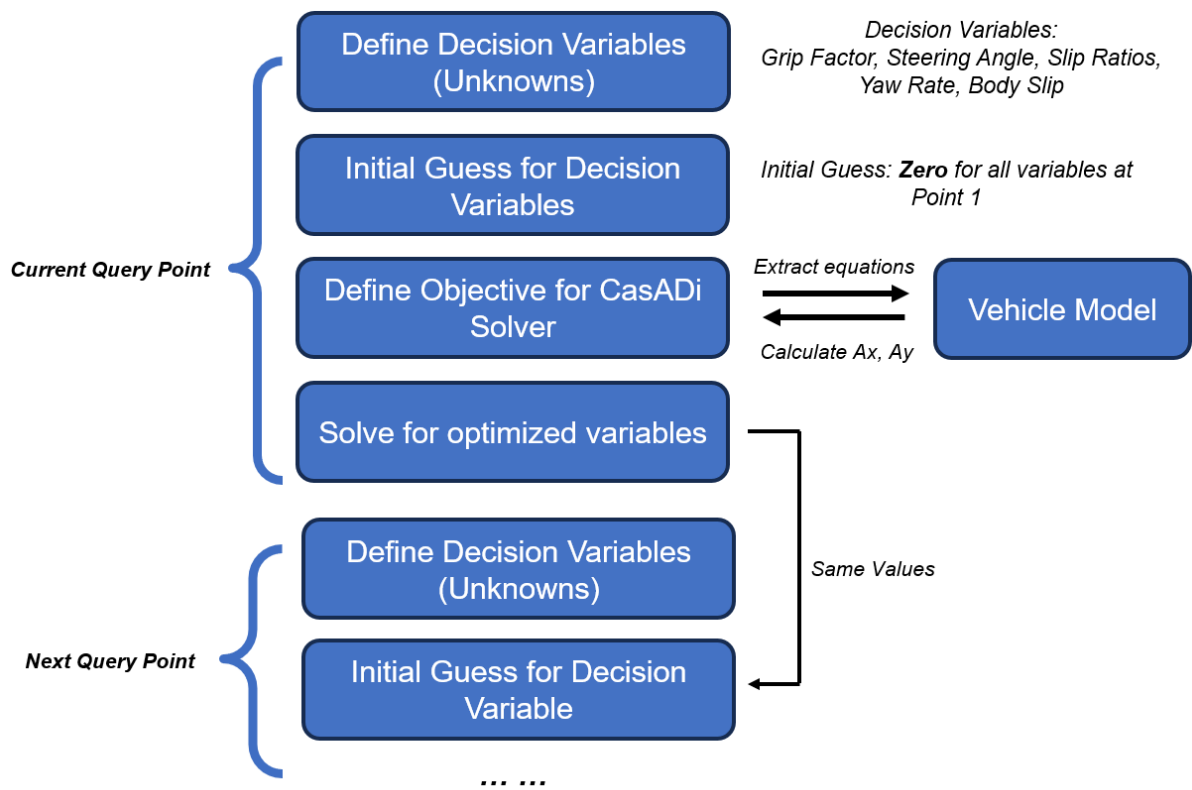


Figure 40 Flow chart of optimization in loop

Several additional design measures were implemented to improve the optimisation success rate. At each query point, the initial guesses for the decision variables are set to the optimised values obtained from the immediately preceding query point, rather than from fixed arbitrary values. Additional measures include the adaptive modification of numerical constraint bounds for certain decision variables within higher speed regimes, where the physical operating conditions deviate significantly from those at lower speeds.

6.2.2 Autocross and Endurance Simulation Validation

Model	Lap Time (s)	Computation Time (s)	Accuracy
R25E	68.36	N/A	100%
LTS25	60.215	232.81	88.09%
LTS26	61.167	132.36	89.48%

Figure 41 Simulation Results Comparison

The simulation results were validated by comparison against telemetry data recorded from the NUS FSAE vehicle of the preceding season, designated R25E, during the Endurance event at the 2025 Formula SAE Electric Competition. To ensure a fair and consistent comparison, the LTS26 simulation was conducted under conditions identical

to those of R25E, and the results were additionally compared against the outputs of the preceding LTS25 model.

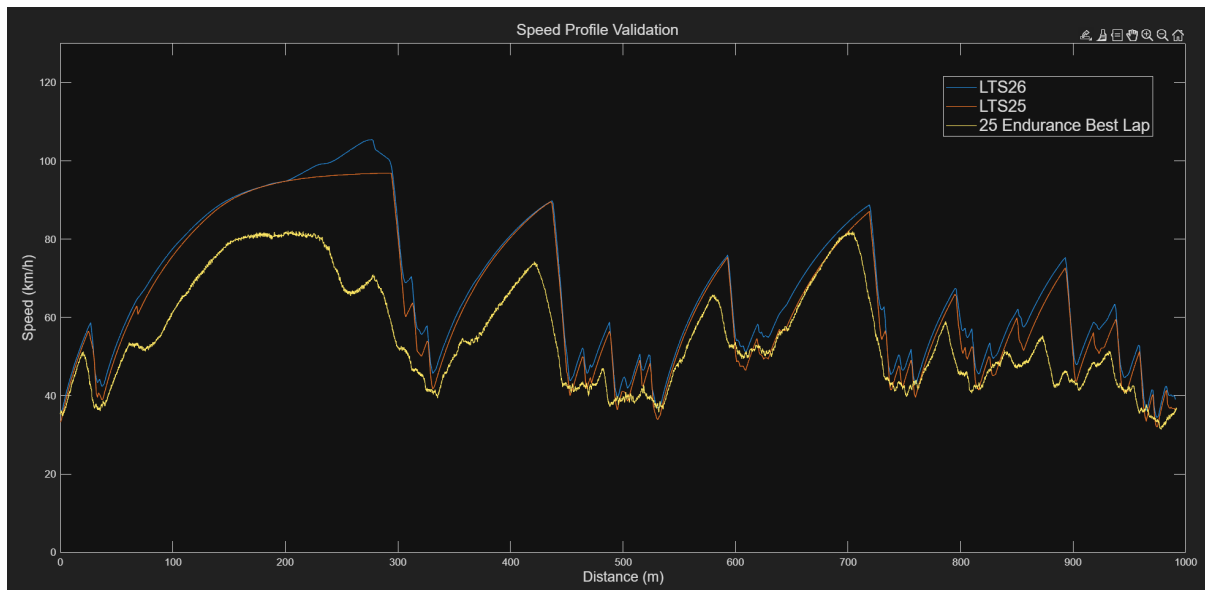


Figure 42 Speed Profile Validation

The simulated speed profile was found to be in close agreement with the competition telemetry data, both in terms of the overall driving pattern and the peak cornering speeds achieved along the track.

Discrepancies between the simulated and recorded speed profiles were observed primarily in the high-speed regions of the run, where the vehicle experienced significant powertrain thermal management constraints resulting in substantial losses in energy transmission efficiency. As the current simulation models the powertrain based on its theoretical torque-speed characteristics without accounting for thermal dissipation effects, it is unable to replicate the performance degradation that occurs under sustained high-load operating conditions in competition.

Furthermore, the simulation was unable to fully replicate the dynamic behaviour of a human driver on track sections such as slaloms, where the turn radius varies rapidly over short distances. This discrepancy reflects an inherent limitation of the quasi-steady-state modelling approach itself, which assumes the vehicle is in equilibrium at each instant and is therefore unable to capture the transient dynamics. Resolving this limitation will ultimately require a transition to a fully transient dynamic simulation framework, which is identified as a long-term development objective in Section 7.

Despite these limitations, LTS26 demonstrated a substantial improvement in computational efficiency relative to its predecessor. As a four-wheel quasi-steady-state model, LTS26 required approximately half the computation time of LTS25, which operated on a simpler two-wheel bicycle model. This improvement not only satisfies the computational efficiency objective established at the outset of the development

programme, but also demonstrates that the more complex four-wheel model can be evaluated more efficiently through the decoupled Performance Envelope architecture introduced in this work. This result further highlights the potential for continued expansion of the vehicle model towards greater fidelity and complexity in future development cycles.

7 Future Work

7.1 Optimization Success Rate

Although a limited number of optimisation failures can be compensated for through interpolation with results from neighbouring query points, an excessive failure rate introduces systematic distortions into the G-G-V diagram and prevents it from accurately representing the vehicle's true performance envelope. The fidelity of the downstream lap simulation is therefore directly contingent on the reliability of the optimisation at each operating point, and reducing the incidence of solver failures represents the most critical priority for the next stage of programme development.

Improving the optimisation success rate is not a one-time fix but an ongoing engineering effort that demands continuous tuning and systematic investigation. In the near term, the primary focus will be on the continued tuning of IPOPT's configurable, which influences convergence behaviour in ways that are highly problem-specific and must be determined empirically through iterative experimentation. In the longer term, the development of a purpose-built optimisation solver tailored to the vehicle dynamics problem formulation can be considered. Sustained and methodical effort across these directions will be essential to progressively reduce the failure rate and improve the overall reliability and accuracy of the simulation.

7.2 Model Complexity and Insights

The current iteration of LTS26 employs a four-wheel vehicle model with three degrees of freedom, encompassing longitudinal motion, lateral motion, and yaw rotation. Future development will focus on progressively expanding the model complexity following the standard hierarchy of vehicle dynamics formulations, from the current three-degree-of-freedom representation, incorporating independent wheel rotational dynamics with 7 DOFs, and ultimately towards a 14-DOFs formulation that accounts for suspension kinematics, roll, pitch, and individual corner load variation.

Beyond model fidelity, a longer-term objective is to transition the core simulation architecture from the current quasi-steady-state framework to a fully transient dynamic model. While the quasi-steady-state approach provides a computationally efficient means of approximating vehicle performance, it inherently assumes that the vehicle is in equilibrium at each instant and cannot capture transient phenomena. A transient model would enable a more physically complete simulation of the vehicle's behaviour throughout a competition lap and providing a stronger foundation for future development of the tool.

Appendix

A.1 Purpose of Lap Time Simulation

Lap Time Simulation (LTS) is a computational modelling tool that predicts the performance of a race car on a given circuit by simulating its motion through the track under the physical constraints imposed by the vehicle's dynamic limits. By numerically solving the equations governing vehicle, the simulation determines the minimum time in which the vehicle can complete a lap, along with the corresponding speed profile, acceleration history, and other performance metrics at every point along the track.

The need for such a tool in a competitive motorsport environment stems from the fundamental challenge that physical testing is inherently limited in scope, cost, and repeatability. Conducting on-track experiments to evaluate every possible combination of vehicle setup parameters is neither time-efficient nor financially viable, particularly for a university-based student team with constrained resources and limited track access. Lap time simulation addresses this challenge directly by enabling the team to evaluate and compare many design and setup configurations in a purely computational environment, at a fraction of the time and cost of physical testing.

Beyond setup optimisation, lap time simulation serves as a powerful design tool during the vehicle development phase. By quantifying the sensitivity of lap time to individual design parameters, the simulation allows engineers to identify which components and systems have the greatest influence on overall performance, enabling design effort and resource allocation to be directed towards the areas of highest impact. This capability is particularly valuable in the context of Formula SAE, where the vehicle is redesigned substantially each season and the team must make critical design decisions under significant time pressure with limited empirical data available.

Furthermore, lap time simulation provides a consistent and objective benchmark against which design changes can be assessed. Unlike physical testing, which is subject to variability introduced by driver behaviour, track conditions, and environmental factors, a simulation produces deterministic results under precisely controlled conditions, making it possible to isolate the effect of a single design variable with a degree of rigour that is practically unachievable through track testing alone. For these reasons, lap time simulation has become an indispensable tool in professional motorsport engineering, and its development and continuous improvement represents a strategic priority for the NUS FSAE Team.

A.2 LTS26 Full Codes

Full work of LTS26 simulation program can be found on GitHub:

[https://github.com/nusfsae/NUS_LTS/tree/main/LTS 26](https://github.com/nusfsae/NUS_LTS/tree/main/LTS%2026)

A.3 LTS26 Variables List

Vehicle Parameters:

Variable	Description
a	Longitudinal distance from the centre of gravity to the front axle
b	Longitudinal distance from the centre of gravity to the rear axle
d	Vehicle track width
delta_in	Steering angle of the inner front wheel
delta_out	Steering angle of the outer front wheel
dx	Longitudinal velocity component in the fixed reference frame
dy	Lateral velocity component in the fixed reference frame
saFR, saFL, saRR, saRL	Tyre slip angles at the front-right, front-left, rear-right, and rear-left corners respectively
Drag	Aerodynamic drag force acting in the direction opposing vehicle motion
Lift	Total aerodynamic downforce acting on the vehicle
AeroF	Aerodynamic downforce distributed to the front axle
AeroR	Aerodynamic downforce distributed to the rear axle
Fz	Static normal load per wheel under unladen conditions
latLT	Lateral load transfer arising from cornering acceleration
longLT	Longitudinal load transfer arising from acceleration or braking
FxFL, FxFR, FxRL, FxRR	Longitudinal tyre forces at the front-left, front-right, rear-left, and rear-right corners respectively
FyFL, FyFR, FyRL, FyRR	Lateral tyre forces at the front-left, front-right, rear-left, and rear-right corners respectively
Mz	Total yaw moment acting about the vehicle centre of gravity
Ppwt	Power demand at the rear wheels required to sustain the current acceleration state
p_inner, p_outer	Polynomial coefficients fitted to the Ackermann steering geometry data for the inner and outer wheel steering angles respectively

Optimization Parameters:

Variable	Description
Gnum	Number of angular subdivisions used to discretise the G-G diagram sweep
V	Vehicle speed at which the G-G diagram is evaluated
GG	Structure of arrays storing the optimised output values across all angular instances
AngleRange	Angular range of the combined acceleration direction, defined from +90° to -90° clockwise
theta	Current angular direction being solved in the G-G diagram sweep
p	Scalar grip factor representing the magnitude of the total acceleration vector — the primary optimisation objective
ax_in	Input longitudinal acceleration component supplied to the vehicle model
ay_in	Input lateral acceleration component supplied to the vehicle model
ax_res	Residual longitudinal acceleration — constrained to zero to ensure consistency between assumed and computed accelerations
ay_res	Residual lateral acceleration — constrained to zero to ensure consistency between assumed and computed accelerations
bias_res	Residual brake bias — constrained to zero during braking to enforce the prescribed front-to-rear braking force ratio
min_Fz	Minimum normal force that each tyre must maintain, enforcing physical contact with the road surface
Ppwt	Powertrain power output constraint, enforcing the vehicle's peak power limit
Mz = 0	Net yaw moment about the centre of gravity constrained to zero, as required for steady-state cornering equilibrium
$ay - V \cdot d\psi = 0$	Lateral acceleration constrained to equal the product of speed and yaw rate, as required for steady-state circular motion
$F_{xfl} \leq 0, F_{xfr} \leq 0$	Front wheel longitudinal forces constrained to be non-positive, reflecting that the front axle is non-driven

A.4 Optimization Solver Selection

The selection of an appropriate numerical optimisation solver is a critical decision, as the solver's algorithmic characteristics directly determine the reliability, computational efficiency, and scalability of the overall simulation framework. Several candidate solvers were evaluated against the requirements of this application, which involves large-scale continuous nonlinear programming with tightly coupled constraints, a non-convex objective landscape arising from the combined-slip tyre model, and the need for repeated solution across a dense grid of operating points.

Solver	Type	Licence	Gradient method	Large-scale performance	Suitability
FMINCON	SQP / Interior point	Free (MATLAB)	Finite difference	Poor — scales badly with problem size	Not suitable
SNOPT	SQP	Commercial	Analytic / finite difference	Good on smooth problems	Limited
KNITRO	SQP / Interior point	Commercial	Analytic	Excellent	Licence Required
Genetic algorithm	Metaheuristic	Free	Derivative-free	Poor — high function evaluation count	Not suitable
IPOPT	Interior point	Free	Analytic (via CasADi)	Excellent	Selected

FMINCON, available natively within MATLAB, was the most immediately accessible candidate. However, its reliance on finite difference approximations for gradient computation introduces numerical noise that degrades convergence reliability on problems with many nonlinear constraints, and its computational performance scales poorly with problem size, rendering it unsuitable for the four-wheel vehicle model considered here.

SNOPT employs a sequential quadratic programming (SQP) approach and exhibits fast local convergence on smooth problems, but requires a feasible or near-feasible initial point to initialise reliably, a condition difficult to guarantee consistently across the full G-G-V operating domain. KNITRO offers robust handling of large-scale non-convex

programmes and would represent a technically strong candidate, but its commercial licensing presents a practical barrier within a student team with limited budget like us.

Derivative-free metaheuristic methods such as genetic algorithms, while capable of exploring non-convex search spaces without gradient information, require a prohibitively large number of function evaluations to converge when the optimisation must be repeated across hundreds of operating points.

IPOPT was selected on several grounds. As an open-source interior point solver designed specifically for large-scale nonlinear optimisation, it imposes no licensing restrictions and integrates directly with CasADi, which supplies exact first and second derivative information through symbolic automatic differentiation, eliminating finite difference noise and enabling efficient exploitation of the sparse problem structure. Relative to the alternatives considered, IPOPT demonstrated superior convergence reliability and computational efficiency for this class of problem. Its extensive configurability further provides the flexibility required to tune solver behaviour as the vehicle model evolves, making it a practical and sustainable long-term foundation for the simulation framework.

A.5 IPOPT Solver Configuration

The following setup for IPOPT was chosen to adapt to the vehicle model, enabling minimal amount of failures and faster computation speeds.

Parameter	Description	Values
opts.print_time	Toggles the output of computation time taken per solver call	False
opts.ipopt.print_level	Controls the verbosity level of IPOPT solver output	1
opts.ipopt.tol	Primary convergence tolerance for the solver; the optimisation is considered converged when the scaled primal and dual infeasibilities fall below this value.	1e-6
opts.ipopt.acceptable_tol	Acceptable convergence tolerance for the solver to consider a solution satisfactory	1e-6
opts.ipopt.acceptable_iter	Number of consecutive successful iterations required before the problem is considered solved	15
opts.ipopt.max_iter	Maximum number of iterations permitted before the solver is forcibly terminated	5000

A.6 Performance Envelope Optimization Setup

In the G-G diagram calculation of LTS26, the program iterates the angle of the vector P to ensure a uniformly meshed ellipse being generated. To facilitate this design, setup for numerical optimization for this iteration loop is as followed.

Vector P being the decision variable that is to be maximized, as it represents the grip level of each iterated instance on the G-G diagram.

Below are constraints to be satisfied to reduce the size of feasible region and speed up the computational speed:

- Steering Angle, Delta, to be within maximum steering angle by design
- Body Slip, Beta, to be maximum 20 degrees as FSAE vehicle is not expected to undergo high body slipping
- Slip ratios, S_x , to be maximum 0.1, as the peak friction provided by tires occurs below 0.1 slip ratios
- Slip angles, S_a , to be maximum 10 degrees, as FSAE vehicle is not expected to have higher tire slip angles
- Yaw rate, $Dpsi$, to be maximum 180 degrees
- Tire loads, F_z , to be always larger than 50N, as the vehicle is expected to be in contact with ground at all time
- $A_y - V \cdot dpsi$, to be always zero, as the vehicle should be balanced at all instance throughout the simulation
- Kinematic Power output from powertrain system, P_{pwt} , to be always less than 80kW, as per rule requires
- Front tires' longitudinal forces, F_{xfl} and F_{xfr} , to be always smaller and equal to zero, as the vehicle is rear-wheel driven

Initial guesses for all decision variables are set to zero initially to ensure the problem is feasible. Subsequent problems take the values of solved decision variables from previous optimization results to provide closer guesses.

Initial guess for grip factor P was set to be 30% of maximum grip value as a conservative starting point, as the vehicle experiences lower downforce and hence tire grips at low speeds. It was reduced to 30% for speeds larger than 20 m/s as a more aggressive guess.

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